Strategic Wishful Thinking: Implications for Forecasts

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Abstract

We characterize how motivated beliefs affect forecasting behavior in the presence of strategic considerations and externalities. Our model provides a unified framework to reconcile recent evidence on the predictability of forecast errors, including under-reaction of consensus forecasts and variation in under- or over-reaction of individual forecasts across different economic environments. Furthermore, we show how the interaction of payoff externalities and wishful thinking can lead forecasters, who are ex-ante symmetric, to endogenously "agree to disagree" about public information: while some choose to underestimate the precision of such information, others overestimate it.

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1 Introduction

Survey forecasts exhibit significant deviations from rational expectations. Using regressions of forecast errors on forecast revisions, Coibion and Gorodnichenko (2012, 2015) show that consensus forecasts exhibit underreaction for macroeconomic variables, while Bordalo, Gennaioli, Ma, and Shleifer (2020) show that individual forecasts tend to exhibit overreaction. In contrast, Bouchaud, Krueger, Landier, and Thesmar (2019) documents underreaction in individual forecasts of financial variables like earnings per share, suggesting that the biases in expectations vary across settings. Ahn and Farmer (2024) finds that forecasters assign heterogeneous weights to public information, and that this source of disagreement is an important source of forecast dispersion, especially during downturns. And Bouchaud et al. (2019) documents that the degree of underreaction varies across forecasters.

Taken together, this evidence raises an important question: why do biases in individual forecasts vary across forecasters and settings? Existing explanations tend to focus on a subset of these patterns.[1] Furthermore, these models typically assume that individuals are symmetric, and so cannot account for differences in the interpretation of public information. While such heterogeneity can be introduced by assuming that forecasters hold different beliefs, such assumptions do not explain when or why such disagreement should arise.

We propose a framework in which individuals, who exhibit "wishful thinking," face uncertainty about the precision of a public signal and choose a subjective interpretation to maximize their anticipatory utility. We characterize how strategic considerations and aggregate externalities affect subjective beliefs and actions. In the context of forecasting, our model implies that individual forecasts tend to exhibit under-reaction (over-reaction) when an individual forecaster is sufficiently worse (better, respectively) off when the consensus forecast is more accurate, reflecting the role of payoff externalities in shaping subjective beliefs. When such payoff externalities are not too extreme, we show that the unique equilibrium can feature *endogenous* disagreement: some individual forecasts exhibit underreaction while the others exhibit overreaction. As such, our analysis sheds light on why individual forecasts exhibit overreaction in some settings, but underreaction in others, and when disagreement about the precision of public information can arise endogenously.

Intuition. We build on the framework of Angeletos and Pavan (2007) in which players'

[1] For instance, models that incorporate diagnostic expectations (e.g., Bordalo et al. (2020)), extrapolation (e.g., Barberis, Greenwood, Jin, and Shleifer (2015)), overconfidence (e.g., Daniel, Hirshleifer, and Subrahmanyam (1998)), or costly information retrieval (e.g., Afrouzi, Kwon, Landier, Ma, and Thesmar (2023)), or account for strategic considerations of forecasters (e.g., Gemmi and Valchev (2023)) focus on generating overreaction in individual forecasts, while models of sticky expectations (e.g., Bouchaud et al. (2019)) generate underreaction. We provide a more comprehensive discussion of these approaches and how our mechanism is different, in Section 5.4.

optimal actions depend not only upon the state of the world (or "fundamentals") but also on strategic considerations and payoff externalities. Our primary application is to the setting of forecasters, and so we describe the intuition of our results in this context. To fix ideas, suppose that forecaster i's payoffs are given by:

$$u_i = -(k_i - \theta)^2 - \rho(k_i - K)^2 - \gamma(K - \theta)^2,$$

where θ denotes the target (forecasted) variable, k_i denotes her forecast, and $K \equiv \int k_i di$ denotes the aggregate or consensus forecast. Specifically, an individual's payoff depends on (i) the accuracy of her forecast, (ii) a preference to either deviate from ($\rho < 0$) or conform to ($\rho > 0$) the consensus forecast, capturing strategic incentives to "stand out" or "herd", and (iii) an independent externality component which reflects the fact that the individual may be better ($\gamma > 0$) or worse ($\gamma < 0$) off when the consensus forecast is more accurate.[2]

Each individual chooses a forecast k_i to maximize the above payoff, given noisy private and public information. While each individual knows the precision of their private information, they face uncertainty about the precision of the public signal.[3] We assume that individuals exhibit "wishful thinking": her choice of precision maximizes her anticipatory utility subject to a penalty from taking suboptimal actions.[4] This implies that strategic considerations and payoff externalities affect an individual's subjective interpretation of information, not just their forecasts.

To see how, first consider a setting in which there are no strategic considerations or externalities (i.e., $\rho = \gamma = 0$). In this case, each individual who engages in wishful thinking has a tendency to over-estimate the precision of the public signal since this leads to higher perceived accuracy and, consequently, higher anticipatory utility - we refer to this as the *fundamental uncertainty channel*. We show that this naturally leads to overreaction in individual forecasts. When individuals' private information is sufficiently precise, this can also generate underreaction in consensus forecasts.[5]

The presence of strategic considerations and externalities generates an additional channel

[2]For equilibrium actions to be well-defined, we need that $\rho \in (-1, 1)$. Note that when $\gamma = 0$, the aggregate action, K, affects an individual's payoff through ρ; however, the nature of this implicit externality is unambiguously linked to the nature of the individual's strategic incentive. We refer to the third term as the externality component because the additional degree of freedom generated by $\gamma \neq 0$ allows us to examine how variation in the externality affects beliefs while holding fixed the impact of an individual's coordination incentives.

[3]In Section 6, we consider the setting where individuals choose their subjective beliefs about the precision of private information.

[4]As we discuss in the next section, we build on the work of Bénabou and Tirole (2016), Brunnermeier and Parker (2005), and most closely, Caplin and Leahy (2019) and Banerjee, Davis, and Gondhi (2024).

[5]Our notions of overreaction and underreaction are as in Coibion and Gorodnichenko (2012) and Bordalo et al. (2020): in regressions of forecast errors on forecast revisions, overreaction (underreaction) refers to the case when the regression coefficient is less than zero (greater than zero, respectively).

through which wishful thinking affects subjective beliefs. For instance, suppose there are no strategic considerations ($\rho = 0$), but there are payoff externalities from aggregate actions ($\gamma \neq 0$). When $\gamma > 0$, errors in the aggregate action (consensus forecast) reduce payoffs, while when $\gamma < 0$, an individual is better off when others make mistakes. Under wishful thinking, this creates incentives for an individual to overestimate the precision of public information when $\gamma > 0$, but underestimate it when $\gamma < 0$. We refer to this as the *aggregate error channel*.[6] Strategic considerations have an analogous effect. When $\rho > 0$, individuals have an incentive to herd, and so the aggregate error channel creates an incentive to overestimate public information precision. In contrast, when $\rho < 0$, individuals prefer to "stand-out" from others, which creates an incentive to underestimate public precision. The overall impact of the aggregate errors channel, then, depends on the interaction of strategic considerations and externalities.[7]

The key question in shaping agents' beliefs is how these two channels interact. For a fixed ρ, when γ is sufficiently high, the fundamental uncertainty and aggregate errors channels reinforce each other, so all individual forecasts exhibit overreaction. In contrast, when γ is sufficiently low, the aggregate errors channel dominates the fundamental uncertainty channel for all individuals, so forecasts exhibit underreaction. In either case, the unique equilibrium is symmetric.

However, for intermediate values of γ, such symmetric equilibria do not exist because neither channel dominates for all individuals. Instead, the unique equilibrium features mixed strategies so that players endogenously choose to "agree to disagree" about the public information: while some individuals underestimate the precision of the public signal, others overestimate it. Intuitively, if an individual believes that many others are overestimating the precision of the public signal, she has a tendency to underestimate the precision, since she is better off believing that others are "wrong". Similarly, if an individual believes that sufficiently many of the others are dismissive of the public information, then she is better off by conditioning on such information. The equilibrium is characterized by the fraction of individuals who over-/under-estimate the public precision that makes every individual indifferent between the two sets of subjective beliefs.

Implications. Our analysis has a number of distinctive implications. First, our model

[6] This distinguishes our analysis from models in which individuals exhibit rational expectations (e.g., Morris and Shin (2002), Angeletos and Pavan (2007), Gemmi and Valchev (2023), where in the absence of strategic considerations (i.e., $\rho = 0$), actions are informationally efficient (i.e., k_i does not exhibit under-reaction or over-reaction). In contrast, wishful thinking implies that even when $\rho = 0$, externalities can distort subjective beliefs and, consequently, equilibrium actions.

[7] The net effect of strategic considerations and externalities is captured by $\chi = -\gamma - \rho - \rho\gamma$, which reflects the net impact of aggregate errors on payoffs. When χ is sufficiently positive (negative), an individual has an incentive to under-estimate (over-estimate) the precision of the public information.

generates underreaction in consensus forecasts and overreaction in individual forecasts when private information is sufficiently precise, even in the absence of strategic considerations and externalities (i.e., $\rho = \gamma = 0$). This distinguishes our mechanism from models in which such results are driven by strategic considerations (e.g., Gemmi and Valchev (2023)).

Second, our analysis implies that, fixing strategic considerations (ρ), individual forecasts should exhibit overreaction when γ is sufficiently high, but *underreaction* when γ is sufficiently low. As we discuss further in Section 5, the latter implies that we should expect to find underreaction for individual forecasts of financial market variables. Intuitively, $\gamma < 0$ captures settings in which individuals are better off when others are wrong. This negative externality is likely to be especially relevant in financial markets, where aggregate mistakes are indicative of mispricing and thus represent profitable trading opportunities (e.g., Bouchaud et al. (2019)). As such, in contrast to many existing explanations, our model provides a common mechanism to explain why we observe overreaction in individual forecasts for some settings, but underreaction in others.

Moreover, we characterize conditions under which individuals endogenously choose to exhibit disagreement about the precision of public information, which distinguishes our results from models in which individuals are exogenously assumed to "agree to disagree." Specifically, we show that endogenous disagreement is more likely when the public information precision is low. This is broadly consistent with Ahn and Farmer (2024), who show that disagreement about public information is more prevalent during economic downturns, to the extent such periods are associated with high uncertainty.

Overview. The next section briefly discusses the related literature. Section 3 presents the general model, a discussion of assumptions, and some preliminary analysis. Section 4 provides a characterization of the optimal subjective beliefs about public information precision in equilibrium. Section 5 applies our analysis to professional forecasts and derives predictions that distinguish our analysis from existing approaches. Section 6 discusses how our analysis extends to subjective beliefs about private signals. We show that wishful thinking leads to (i) overestimating the precision of one's own signal i.e., overconfidence, but (ii) subjective beliefs about others' signals depend on the relative impact on payoffs of aggregate volatility versus dispersion in others' actions. Section 7 concludes.

2 Related Literature

The concept of anticipatory utility, or current subjective expected utility, dates to at least Jevons (1905).[8] Specifically, anticipation of a future experience can directly impact an

[8] We utilize the term anticipatory utility as distinct from the concept of *anticipated* utility, whereby agents hold fixed the parameters they learn about when choosing actions (e.g., Kreps (1998), Cogley and Sargent

individual's current utility (e.g., anxiety about a risky medical procedure, excitement about an upcoming vacation). There is now an extensive literature that incorporates anticipatory utility into models of belief choice (e.g., Akerlof and Dickens (1982), Loewenstein (1987), Caplin and Leahy (2001)).[9]

The most closely related papers are Brunnermeier and Parker (2005), Caplin and Leahy (2019) and Banerjee et al. (2024). Brunnermeier and Parker (2005) show how subjective belief choice ("optimal expectations") can help explain a preference for skewness, portfolio under-diversification, and consumption/savings patterns. Caplin and Leahy (2019) shows how wishful thinking can generate a large number of common behavioral biases, including confirmation bias and procrastination. Consistent with this idea, a large experimental and field literature documents that beliefs respond systematically to incentives, with individuals holding more optimistic or pessimistic beliefs when doing so improves expected payoffs or future opportunities (e.g., Möbius, Niederle, Niehaus, and Rosenblat, 2022; Bénabou and Tirole, 2016; and Coutts, 2019).

Our analysis builds on this earlier work, but focuses on the impact of the behavior of others (via strategic considerations and externalities) on subjective belief choice. Importantly, we show this can give rise to endogenous disagreement about the interpretation of public information. Banerjee et al. (2024) explores how wishful thinking affects investors' interpretation of endogenous price information in a stylized financial market setting. In contrast, the current paper focuses on the interpretation of *exogenous* public information in a more general setting, featuring both strategic substitutability and complementarity as well as aggregate externalities.

We adopt the framework of Angeletos and Pavan (2007), which generalizes the canonical "beauty contest" model of Morris and Shin (2002) to a large class of quadratic-Gaussian economies. As such, our paper is related to the rich literature on coordination games that analyzes how strategic considerations affect individuals' information acquisition and attention decisions (e.g., Hellwig and Veldkamp, 2009; Myatt and Wallace, 2012; Colombo, Femminis, and Pavan, 2014; Hébert and La'O, 2020). Recent work has extended this class of quadratic coordination models by endogenizing information structures. Pavan (2025) studies how agents endogenously allocate attention to private information in coordination games with bounded recall, showing how strategic considerations shape equilibrium information acquisition. Complementarily, Inostroza and Pavan (2025) analyze the endogenous design

(2008)).

[9] Importantly, individuals do not exhibit "multiple selves" but consciously hold a single set of beliefs about the world so that an individual's subjective beliefs affect her actions, too. As Bénabou and Tirole (2016) emphasize, this generates tension between holding "accurate" beliefs, which lead to ex-post optimal actions, and "desirable" beliefs, which increase contemporaneous utility.

of public information in coordination environments, characterizing optimal public signal structures under adversarial objectives. In contrast, our analysis takes the information structure as given and focuses on how strategic considerations and payoff externalities shape agents' subjective interpretation of public information.

A natural parallel to "wishful thinking" is ambiguity aversion and robust control (e.g., Hansen and Sargent (2001, 2008)) in which an individual chooses action a and subjective beliefs μ to solve

$$\min_{\mu \in [\underline{\mu}, \bar{\mu}]} \max_{a} \mathbb{E}_{\mu} \left[u(a) \right],$$

where $\mathbb{E}_{\mu} \left[u(a) \right]$ reflects the subjective expected utility from action a under "worst case" beliefs, where μ is chosen from a set of reasonable beliefs $\left[\underline{\mu}, \bar{\mu}\right]$.[10] In our setting, players solve an analogous problem: they choose action a and beliefs μ to solve:

$$\max_{\mu \in [\underline{\mu}, \bar{\mu}]} \max_{a} \mathbb{E}_{\mu} \left[u(a) \right].$$

In this case, the optimal μ reflects the "wishful thinking" that the agent engages in to increase anticipatory utility $\mathbb{E}_{\mu} \left[u(a) \right]$ – in a sense, the agent chooses actions that perform well given "best case" outcomes.[11] While both wishful thinking and ambiguity aversion are likely to be relevant to how individuals interpret information in strategic situations, we leave a model that incorporates both approaches for future work.

3 Model

3.1 Setup

Our analysis builds on the generalized setting formalized in Angeletos and Pavan (2007). There is a unit measure continuum of players indexed by $i \in [0, 1]$. Each player chooses an action, $k_i \in \mathbb{R}$, to maximize her expected payoff, U_i, which also depends upon the true state of the world, θ, as well as the actions of all other players, denoted by the vector k_{-i}. We assume that U_i is (i) quadratic in its arguments and (ii) symmetric across the actions of other players (i.e., $U_i(k_i, k_{-i}, \theta) = U_i\left(k_i, k'_{-i}, \theta\right)$ for any permutation k'_{-i} of k_{-i}). Let $K \equiv \int_0^1 k_j dj$ denote the average action of all other players and $\sigma_k \equiv \left(\int_0^1 (k_j - K)^2 dj\right)^{\frac{1}{2}}$ denote the dispersion of others' actions.

As Angeletos and Pavan (2007) show, the above implies that payoffs can be expressed as

[10] For example, Dupraz (2015) considers the impact of such preferences in the canonical beauty contest model of Morris and Shin (2002))

[11] It is worth noting that an alternate interpretation of robust control is that the agent's beliefs coincide with the objective distribution, but she faces uncertainty about these beliefs and chooses actions that perform well given adverse outcomes. Analogously, one could interpret the choices of a "wishful thinker" as actions that perform well given favorable outcomes.

a function $U_i \equiv u(k_i, K, \sigma_k, \theta)$, where $u(\cdot)$ is quadratic and its partials satisfy $u_{k\sigma} = u_{K\sigma} = u_{\theta\sigma} = 0$ and $u_\sigma(k, K, 0, \theta) = 0$ for all (k, K, θ). This generalized functional form ensures tractability while still preserving flexibility for our analysis.[12] Importantly, it allows for both externalities from aggregate activity ($u_K \neq 0$ or $u_\sigma \neq 0$) and strategic substitutability or complementarity ($u_{kK} < 0$ or $u_{kK} > 0$, respectively) in actions. The following assumption ensures that equilibrium actions are unique and bounded.

Assumption 1. *$u(k_i, K, \sigma_K, \theta)$ satisfies both (i) $u_{kk} < 0$, and (ii) $r \equiv -u_{kK}/u_{kk} < 1$.*

Equilibrium actions depend upon fundamentals, i.e., $u_{k\theta} \neq 0$, but players have incomplete information when choosing their actions. Specifically, $\theta \sim N(0, 1/\tau)$ and each player observes both a private signal s_i and a public signal s, where

$$s_i = \theta + \varepsilon_i, \quad \text{and} \quad s = \theta + \eta, \tag{1}$$

where the common error has distribution $\eta \sim N(0, 1/\tau_\eta)$, the individual errors have distribution $\varepsilon_i \sim N(0, 1/\tau_e)$, and all random variables $\{\theta, \eta, \{\varepsilon_i\}_i\}$ are mutually independent.

We assume players face uncertainty about the quality of public signal and allow them to entertain subjective beliefs about the precision of such information.[13] Specifically, we allow player i to perceive the error in the public signal to be:

$$\eta \sim_i N\left(0, \frac{1}{\delta_{\eta,i}\tau_\eta}\right). \tag{2}$$

If $\delta_{\eta,i} = 1$, player i's beliefs coincide with the objective distribution: she exhibits **rational expectations**. When $\delta_{\eta,i}$ is greater (lesser) than one, player i overestimates (underestimates) the precision of the public signal. We assume that players entertain subjective beliefs that allow for both over-estimation and under-estimation, but that $\delta_{\eta,i}$ are bounded below to ensure that they cannot completely ignore the public signal, i.e.,

Assumption 2. *There exists $0 < \underline{\delta} < 1$ and $1 < \bar{\delta} \leq \infty$, such that $\underline{\delta} \leq \delta_{\eta,i} \leq \bar{\delta}$ for all i.*

In all other ways, the players are rational: in particular, they (i) take as given other players' actions and beliefs and (ii) update using Bayes' rule. For ease of notation, we denote the expectation and variance of the random variable X, given player i's subjective beliefs $\delta_{\eta,i}$, by $\mathbb{E}_i[X]$ and $\text{var}_i[X]$, respectively. Each player's optimal action, $k_i^*(\delta_{\eta,i})$, maximizes her expected payoff, given her subjective beliefs, i.e.,

$$k_i^*(s_i, s; \delta_{\eta,i}) \equiv \arg\max_{k_i} \mathbb{E}_i\left[u(k_i, K, \sigma_k, \theta) \,\middle|\, s_i, s\right], \tag{3}$$

[12] As noted in Angeletos and Pavan (2007), one can also interpret this setting as a second-order approximation of a much broader class of economies.

[13] In Section 6, we allow players to entertain subjective beliefs about their own private signal. We also study the implications when players hold subjective beliefs about other players' private information. The joint distribution of beliefs analyzed in the benchmark model can be found in equation (31).

where we suppress the dependence on $\{s_i, s\}$ when convenient. Given this optimal action, we denote player i's **anticipatory utility**, i.e., the contemporaneous utility flow she receives from her beliefs about her expected payoff, by:

$$AU_i\left(\delta_{\eta,i}\right) \equiv \mathbb{E}_i\left[u\left(k_i^*\left(s_i, s; \delta_{\eta,i}\right), K, \sigma_k, \theta\right)\right], \tag{4}$$

whereas her expected **experienced utility**, given her chosen action, is

$$EU_i\left(\delta_{\eta,i}\right) = \mathbb{E}\left[u\left(k_i^*\left(s_i, s; \delta_{\eta,i}\right), K, \sigma_k, \theta\right)\right]. \tag{5}$$

Each player chooses her subjective beliefs to maximize a weighted average of her anticipatory utility and her experienced utility. Formally, player i's objective function is

$$\max_{\delta_{\eta,i}\in[\underline{\delta},\bar{\delta}]} \frac{1}{1+\psi}\left\{AU_i(\delta_{\eta,i}) + \psi EU_i(\delta_{\eta,i})\right\} \equiv TU_i(\delta_{\eta,i}), \tag{6}$$

where $\psi \geq 0$ scales the relative importance of experienced utility versus anticipatory utility.

The specification of the objective in (6) highlights the tradeoff that players face between "desirable" models (that increase anticipatory utility) against "accurate" models (that increase experienced utility) in a transparent and tractable manner.[14] When $\psi \to 0$, the player ignores the impact of her subjective belief distortion on her experienced utility - this leads to a pure focus on "desirable" models and maximal belief distortion within the set of reasonable beliefs. On the other hand, when $\psi \to \infty$, the player effectively only cares about the experienced utility, and the optimal choice of beliefs converges to the rational expectations benchmark i.e., $\delta_{\eta,i} = 1$.

The notion of equilibrium in our setting is as follows:

Definition 1. *An* **equilibrium** *consists of actions $\{k_i^*\}_i$ and subjective beliefs $\{\delta_{\eta,i}\}_i$ such that (i) player i's action $k_i^*(s_i, s; \delta_{\eta,i})$ maximizes (3) given her subjective beliefs, (ii) player i's subjective belief choice $\delta_{\eta,i}$ maximizes (6), and (iii) player i's subjective beliefs are consistent with the equilibrium choices of other players and satisfy Bayes rule, given subjective beliefs. A* **linear equilibrium** *is an equilibrium in which each individual's actions $k_i^*(s_i, s; \delta_{\eta,i})$ are linear in her private and public signals $\{s_i, s\}$.*

In what follows, we distinguish between two types of equilibria: (i) a pure strategy, or symmetric, equilibrium in which all players have the same interpretation of the public signal (i.e., $\delta_{\eta,i}$ is the same), and (ii) a mixed strategy, or asymmetric, equilibrium in which players disagree about the interpretation of the public signal (i.e., the equilibrium $\delta_{\eta,i}$ differs across players).

[14] This is related to the specification of "psychological expected utility" in Caplin and Leahy (2001) and Caplin and Leahy (2004).

3.2 Discussion of Assumptions

As we shall see, Assumption 2 implies that our problem is well-behaved by ensuring that an individual cannot achieve infinite marginal (anticipatory) utility by believing that the public signal is infinitely noisy (i.e., $\delta_{\eta,i} = 0$). Moreover, we require that the upper bound $\bar{\delta} > 1$ to ensure that individuals can choose to overestimate the precision of the public signal. While the lower bound is a necessary technical assumption for our analysis, a natural interpretation of such restrictions on the set of beliefs is that they arise from the confidence intervals associated with their estimates of the joint distribution of fundamentals and signals.[15] We follow Brunnermeier and Parker (2005) by assuming that the cost of deviating from rational expectations is given by the loss in experienced utility due to suboptimal actions. An alternative approach would be to assume the cost of choosing subjective beliefs is dependent upon a measure of statistical distance (e.g., the K-L distance), as in Caplin and Leahy (2019).[16] We expect our results would be qualitatively similar under this alternate specification, but the analysis is less tractable.

We restrict attention to subjective belief choice about public information for tractability and expositional clarity. In Section 6, we also allow players to choose their subjective beliefs about the precision of their own private signals and those of others. In Appendix B.3, we characterize how players choose subjective beliefs about the means of each signal. While players never distort their beliefs about their own private signal, we show that across many applied settings, players may endogenously disagree about the mean of the public signal. In principle, we could have allowed for a player to exhibit wishful thinking about the behavior of others' reactions. However, this involves allowing players to form subjective beliefs about endogenous equilibrium objects, which is conceptually more challenging and makes the analysis less tractable; as such, we leave this for future work.

3.3 Equilibrium actions

We begin by first characterizing the equilibrium actions given an arbitrary choice of subjective beliefs. If all players observed θ perfectly, the optimal action would be

$$\kappa(\theta) \equiv \kappa_0 + \kappa_1\theta, \quad \text{where} \quad \kappa_0 \equiv -\frac{u_k(0,0,0,0)}{u_{kk} + u_{kK}}, \quad \kappa_1 \equiv -\frac{u_{k\theta}}{u_{kk} + u_{kK}}. \tag{7}$$

[15] In their survey, Epley and Gilovich (2016) provide supportive evidence: individuals form a single, subjective model of the world through motivated reasoning, but it must be "reasonable", i.e., this model is naturally bound by the limits of the observable evidence.

[16] As Caplin and Leahy (2019) point out, these different approaches do not require players to know, or believe, the objective distribution but simply that they find it difficult to hold beliefs that deviate too far from it. This may be because such deviations lead to suboptimal decisions (in terms of experienced utility, as we model it), or because it is more difficult to empirically collect supportive evidence for such beliefs, or because they have a preference for social conformity.

Note that, given the specification of payoffs, u_{kk}, u_{kK}, $u_{k\theta}$ are constants. However, players have incomplete information. Given player $i's$ subjective beliefs and the realization of s_i and s, her optimal action, k_i is

$$k_i^*\left(s_i, s; \delta_{\eta,i}\right) = r\mathbb{E}_i\left[K|s_i, s\right] + (1-r)\,\mathbb{E}_i\left[\kappa\left(\theta\right)|s_i, s\right], \tag{8}$$

where $r \equiv -\frac{u_{kK}}{u_{kk}}$ is the **equilibrium degree of coordination** across players. This term captures the extent to which each player chooses to align their action with their expectation of others' actions, K, relative to her expectation of the full-information target, $\kappa\left(\theta\right)$. Assumption 1 ensures that r is bounded above, which ensures the existence and uniqueness of a linear equilibrium.

Given our assumptions on the joint distribution of fundamentals and signals, Bayesian updating implies that player i's conditional beliefs about θ are given by

$$\mathbb{E}_i\left[\theta|s_i, s\right] = A_i s_i + B_i s, \text{ and } \mathrm{var}_i\left[\theta|s_i, s\right] = \frac{1 - A_i - B_i}{\tau}, \tag{9}$$

where player i's weights on the private and public signals are given by:

$$A_i \equiv \frac{\tau_e}{\tau + \tau_e + \delta_{\eta,i}\tau_\eta}, \text{ and } B_i \equiv \frac{\delta_{\eta,i}\tau_\eta}{\tau + \tau_e + \delta_{\eta,i}\tau_\eta}. \tag{10}$$

We define the aggregate weight on private information, $A \equiv \int_i A_i di$, and the public information, $B \equiv \int_i B_i di$, analogously. The following lemma characterizes equilibrium actions.

Lemma 1. *Given a choice of subjective beliefs $\{\delta_{\eta,i}\}_i$ for each player i, a linear equilibrium exists, is unique, and is characterized by $k_i\left(s_i, s; \delta_{\eta,i}\right) = \kappa_0 + \alpha_i s_i + \beta_i s$ and $K = \kappa_0 + \alpha\theta + \beta s$, where $\alpha_i = \frac{1-r}{1-rA}A_i\kappa_1$, $\beta_i = \frac{(1-r)B_i + Br}{1-rA}\kappa_1$, $\alpha = \frac{A(1-r)}{1-rA}\kappa_1$, and $\beta = \frac{B}{1-rA}\kappa_1$.*

While the form of this solution is standard, the distinguishing feature of our setting is that the relative weights placed on the private and public signal (α_i and β_i, respectively) are distorted by each player i's subjective beliefs (through A_i and B_i). When there is a benefit to coordination, i.e., if $r \neq 0$, these weights also depend upon the subjective beliefs of *other* players through the aggregate weights on private and public information (A and B, respectively) which player i takes as given. We show below that both dimensions play a critical role in the equilibrium choice of beliefs.

4 Optimal subjective beliefs

This section presents our main analysis. We first characterize the key trade-off that a player faces when choosing her subjective beliefs (Section 4.1). To build intuition, we first solve the model when players maximize anticipatory utility only, i.e., when $\psi = 0$ (Section 4.2). In Section 4.3, we then characterize the equilibrium in the general setting, i.e., when $\psi \neq 0$.

4.1 Relative impact of aggregate errors

We begin with a characterization of player i's anticipatory utility, which depends on her beliefs about both fundamentals and the equilibrium actions of others.

Lemma 2. *Given player i's subjective beliefs, anticipatory utility can expressed as:*

$$AU_i\left(\delta_{\eta,i}\right) = \left(\frac{\kappa_1}{1-rA}\right)^2 \left[\underbrace{u_{kk}\left(1-r\right)^2 var_i\left[\theta|s_i,s\right]}_{\textit{fundamental uncertainty channel}} + \underbrace{\left(u_{KK}-r^2u_{kk}\right)B^2 var_i\left[s|\theta\right]}_{\textit{aggregate errors channel}}\right] + \bar{\mathcal{A}}_i, \quad (11)$$

where $\bar{\mathcal{A}}_i$ does not depend on player i's subjective beliefs ($\delta_{\eta,i}$).

A player's subjective beliefs affect her anticipatory utility through two channels. First, when the player believes that her fundamental forecast is more precise, i.e., when $\text{var}_i\left[\theta|s_i,s\right]$ decreases, her anticipatory utility increases (since $u_{kk} < 0$ in equation (11)) - we refer to this as the **fundamental uncertainty channel**. Second, the player's anticipatory utility is affected by her beliefs about the excess volatility in the aggregate action K, which is driven by $B^2\text{var}_i[s|\theta]$, since

$$\text{var}_i[K|\theta] = \left(\frac{\kappa_1}{1-rA}\right)^2 B^2\text{var}_i[s|\theta],$$

where $\text{var}_i[s|\theta] = \frac{1}{\delta_{\eta,i}\tau_\eta}$. We refer to this as the **aggregate errors channel**, because it captures the impact of "errors" in the aggregate action (relative to the full information case).[17]

The aggregate errors channel highlights a novel mechanism through which wishful thinking induces distortions in a player's subjective beliefs. To see why, note that one can express[18]

$$\frac{\partial AU_i}{\partial \delta_{\eta,i}} \propto \left(-\frac{\partial \text{var}_i\left[\theta|s_i,s\right]}{\partial \delta_{\eta,i}} + \chi B^2 \frac{\partial \text{var}_i\left[s|\theta\right]}{\partial \delta_{\eta,i}}\right) = \frac{\tau_\eta}{\left(\tau+\tau_e+\delta_{\eta,i}\tau_\eta\right)^2} - \frac{\chi B^2}{\delta_{\eta,i}^2\tau_\eta}, \quad (12)$$

where the **relative impact of aggregate errors** is captured by

$$\chi \equiv \frac{u_{KK}-r^2u_{kk}}{-u_{kk}\left(1-r\right)^2} = \frac{r^2-\frac{u_{KK}}{u_{kk}}}{\left(1-r\right)^2}. \quad (13)$$

To gain some intuition for how χ depends on externalities (u_{KK}) and strategic considerations (r), first consider the case where $r=0$, so that χ has the same sign as u_{KK}. Because there are no strategic considerations, the impact of subjective beliefs $\delta_{\eta,i}$ is completely determined

[17] Notably, since players have objective beliefs about the private information of others, subjective beliefs do not affect anticipatory utility via their impact on the dispersion of others' actions (i.e., $u_{\sigma\sigma}$). In Section 6, we relax this assumption.

[18] Equation (12) highlights the importance of Assumption 2 which imposes bounds on $\delta_{\eta,i}$: otherwise, one could ensure infinite marginal utility when $\chi < 0$ by setting $\delta_{\eta,i} = 0$.

by how the volatility of aggregate actions directly affects payoffs (i.e., u_{KK}). When $u_{KK} > 0$ (so that $\chi > 0$), the aggregate errors channel decreases the marginal anticipatory utility: since the player is better off when aggregate action is more volatile, she has an incentive to underestimate the precision of the public signal. Similarly, when $u_{KK} < 0$, the aggregate errors channel implies that a player has an incentive to overestimate public precision.

Next, to understand the impact of strategic considerations, consider the special case where:

$$u_i = -(\theta - k_i)^2 - \rho(k_i - K)^2. \tag{14}$$

In this case, note that $r = \frac{\rho}{1+\rho}$ and $-\frac{u_{KK}}{u_{kk}} = -\frac{\rho}{1+\rho}$, and so the relative impact of aggregate errors is $\chi = -\rho$.[19] In this case, the aggregate errors channel implies that players have an incentive to overestimate (underestimate) the precision of the public signal when actions are strategic complements (substitutes, respectively). It is important to note, however, that more generally, the sign of χ does not depend upon whether players' actions are strategic complements (i.e., $r > 0$) or substitutes ($r < 0$), but on the net impact of strategic considerations and externalities. In fact, as we will highlight in Section 5, χ can be positive or negative even when there are no strategic considerations (i.e., when $r = 0$).

The overall impact of subjective beliefs about public precision on anticipatory utility then depends on the sign and relative magnitude of χ. When $\chi < 0$, anticipatory utility decreases in the volatility of aggregate errors so an increase in $\delta_{\eta,i}$ unambiguously increases anticipatory utility. When $\chi > 0$, the fundamental uncertainty and aggregate errors channels operate in opposite directions. If $\delta_{\eta,i}$ is sufficiently low or χB^2 is sufficiently high, the aggregate errors channel dominates and so anticipatory utility decreases with $\delta_{\eta,i}$. When $\delta_{\eta,i}$ is sufficiently high or χB^2 is sufficiently low, the informational channel dominates and so anticipatory utility increases with $\delta_{\eta,i}$. As we shall see next, this can give rise to endogenous disagreement for intermediate values of χ.

4.2 Benchmark: Maximizing Anticipatory Utility

In this section, we characterize the equilibrium when $\psi = 0$ i.e., when players ignore the impact of their chosen beliefs on experienced utility. Let

$$\underline{B} = \frac{\underline{\delta}\tau_\eta}{\tau + \tau_e + \underline{\delta}\tau_\eta}, \quad \bar{B} = \frac{\bar{\delta}\tau_\eta}{\tau + \tau_e + \bar{\delta}\tau_\eta}, \quad \text{and} \quad B_{RE} = \frac{\tau_\eta}{\tau + \tau_e + \tau_\eta}. \tag{15}$$

A player puts weight $\bar{B}$, $\underline{B}$ and B_{RE} on the public signal if she chooses $\delta_{\eta,i} = \bar{\delta}$ (she maximally overestimates the precision of public information), $\delta_{\eta,i} = \underline{\delta}$ (she maximally under-estimates the precision) or $\delta_{\eta,i} = 1$ (she exhibit rational expectations), respectively.

[19] This example also illustrates that in many applied specifications for which $\gamma = 0$, r and u_{KK} are often tightly linked.

We begin by characterizing the optimal belief choice of player i when all other players exhibit rational expectations.

Corollary 1. *Suppose $\psi = 0$ and all other players exhibit rational expectations i.e., for all $j \neq i$, $\delta_{\eta,j} = 1$. Then, player i chooses $\delta_{\eta_i} = \underline{\delta}$ if $\chi > \frac{\bar{B}\underline{B}}{B_{RE}^2}$, $\delta_{\eta,i} = \bar{\delta}$ if $\chi < \frac{\bar{B}\underline{B}}{B_{RE}^2}$, and is indifferent otherwise.*

As we establish in the proof, even if all other players exhibit rational expectations, an individual player finds it optimal to deviate and exhibit wishful thinking. When χ is sufficiently high, the aggregate errors channel dominates and the player chooses to believe that the public signal is noisier than it actually is; otherwise, she over-estimates the precision of the public signal. This intuition carries over into the equilibrium in which *all* players choose their beliefs to maximize anticipatory utility.

Proposition 1. *Suppose $\psi = 0$ and all players optimally choose their subjective beliefs.*

(i) If $\chi > \bar{B}/\underline{B}$, then a unique equilibrium exists, and is such that $\delta_{\eta,i} = \underline{\delta}$ for all i.

(ii) If $\chi < \underline{B}/\bar{B}$, then a unique equilibrium exists, and is such that $\delta_{\eta,i} = \bar{\delta}$ for all i.

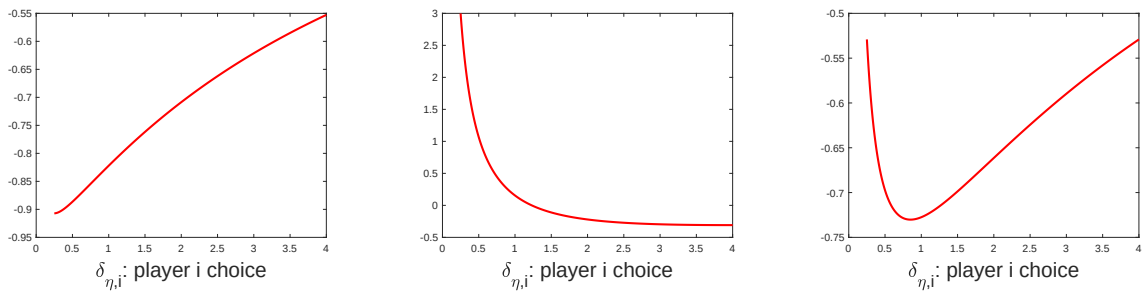
(iii) If $\chi \in \left[\underline{B}/\bar{B},\, \bar{B}/\underline{B}\right]$, then a unique equilibrium exists, and is such that a fraction λ of players choose $\delta_{\eta,i} = \bar{\delta}$ while all others choose $\delta_{\eta,i} = \underline{\delta}$, where

$$\lambda = \frac{\sqrt{\frac{\bar{B}\underline{B}}{\chi}} - \underline{B}}{\bar{B} - \underline{B}}, \tag{16}$$

As in Corollary 1, it is always optimal to over-estimate (under-estimate) the precision of public information when χ is sufficiently low (high). However, when χ is in an intermediate region, there cannot exist a pure-strategy, symmetric equilibrium in which all players interpret the public information in the same way. This is due to the impact that the equilibrium beliefs of the other players have on player i's incentives. For instance, if all other players choose to under-react to the public signal so that $B = \underline{B} < B_{RE}$, then player i's incentive to choose $\underline{\delta}$ falls: the aggregate errors channel is less important, as seen in (12). If $\chi > \underline{B}/\bar{B}$, player i has an incentive to deviate and chooses to over-react (i.e., choose $\delta_{\eta,i} = \bar{\delta}$).

As a result, when $\chi \in \left[\underline{B}/\bar{B},\, \bar{B}/\underline{B}\right]$, the unique equilibrium features *endogenous disagreement* about the interpretation of the public signal: a fraction λ of players choose to over-react to the public signal, while the remaining players under-react to it. This optimal λ ensures that the relative impact of the aggregate errors and fundamental uncertainty channels exactly offset so that each player is indifferent between the two choices: as χ increases, λ

Figure 1: Anticipatory utility versus $\delta_{\eta,i}$ when $\chi \in \left[\underline{B}/\bar{B},\ \bar{B}/\underline{B}\right]$
The figure plots the anticipatory utility for player i as a function of $\delta_{\eta,i}$. Other parameters: $\tau = \tau_e = \tau_\eta = 1$, $\chi = 1$, $\psi = 0$ and $\bar{\delta} = 1/\underline{\delta} = 4$.



(a) $B = \underline{B}$ (b) $B = \bar{B}$ (c) Mixed eqm.

and, as a result, B decreases so that χB^2 stays constant in this interval.[20] Intuitively, as χ increases, the measure of players who choose $\underline{\delta}$ increases, reflecting the increased importance of the aggregate errors channel.

Figure 1 provides a numerical illustration of this intuition. The panels show player i's anticipatory utility as a function of $\delta_{\eta,i}$, given the beliefs of others. In panel (a), all other agents choose $\delta_\eta = \underline{\delta} = 1/4$. Since $B = \underline{B}$ is sufficiently low, the information channel dominates and player i deviates by over-weighting the public signal (chooses $\delta_{\eta,i} = 4$). In panel (b), all other agents choose $\delta_\eta = \bar{\delta} = 4$. Now, the aggregate errors channel dominates, i.e., $B = \bar{B}$ is sufficiently large, and player i strictly prefers to believe the public signal is (relatively) uninformative (chooses $\delta_{\eta,i} = 1/4$). In both cases, a symmetric equilibrium is ruled out. Panel (c) illustrates the mixed strategy equilibrium in which a fraction $\lambda = 0.25$ of players overweight the public signal while the remaining fraction $1-\lambda = 0.75$ underweight it. As is clear, given the choice of all other agents, player i is indifferent between these two (sets of) beliefs.

Next, we characterize how the range of χ for which a mixed equilibrium obtains depends on parameters.

Corollary 2. *The ratio $\bar{B}/\underline{B}$, and consequently, the region $\chi \in \left[\underline{B}/\bar{B},\ \bar{B}/\underline{B}\right]$, is increasing in τ and τ_e, but decreasing in τ_η.*

The above result suggests that, all else equal, players are more likely to endogenously disagree about public information when prior variance about fundamentals is low, when private information is precise, and when public information is noisy. Crucially, these

[20] The equilibrium B is a continuous function of the underlying parameters: if $\chi = \underline{B}/\bar{B}$, $\lambda = 1$, consistent with the symmetric equilibrium that obtains when $\chi < \underline{B}/\bar{B}$; if $\chi = \bar{B}/\underline{B}$, $\lambda = 0$ consistent with the symmetric equilibrium that obtains when $\chi > \bar{B}/\underline{B}$.

predictions distinguish our analysis from models in which agents are (exogenously) assumed to disagree about public information.

4.3 Trading off anticipatory versus experienced utility

The analysis in Section 4.2 highlights the economic channels through which subjective belief choice can endogenously give rise to disagreement about public information. However, when players maximize their anticipatory utility only ($\psi = 0$), their belief choices are extreme within the set of "reasonable beliefs". When $\psi > 0$, each player faces an additional trade-off: over- or under-weighting the public signal reduces the informational efficiency of her action, reducing her experienced utility. The following proposition characterizes equilibria in this setting.

Proposition 2. *Let* $B_{opt} = \max\left\{\min\left\{\frac{B_{RE}(2\psi+1)-\psi\underline{B}}{2\psi}, \bar{B}\right\}, \underline{B}\right\}$ *and* $\Gamma = 1 + \psi\left(2 - \frac{\underline{B}+B_{opt}}{B_{RE}}\right) > 0$.

(i) If $\chi > \frac{B_{opt}}{\underline{B}}\Gamma$*, then a unique equilibrium exists, and is such that* $\delta_{\eta,i} = \underline{\delta}$ *for all* i*.*

(ii) If $\chi < \frac{\underline{B}}{B_{opt}}\Gamma$*, then a unique equilibrium exists, and is such that, for all* i

$$\delta_{\eta,i} = \min\left\{\bar{\delta}, \frac{2\psi + 1 - \chi}{2\psi - \frac{(1-\chi)\tau_\eta}{\tau_e+\tau}}\right\} \equiv \delta_{sym}. \tag{17}$$

In particular, if $\chi = 0$*, then for all* i*,* $\delta_{\eta,i} = \min\{\bar{\delta}, \frac{2\psi+1}{2\psi-\frac{\tau_\eta}{\tau_e+\tau}}\} > 1$*.*

(iii) If $\chi \in \left[\frac{\underline{B}}{B_{opt}}\Gamma, \frac{B_{opt}}{\underline{B}}\Gamma\right]$*, then a unique equilibrium exists, and is such that a fraction* λ *of players choose* $\delta_{\eta,i} = \delta_{opt}$*, while all others choose* $\delta_{\eta,i} = \underline{\delta}$*, where*

$$\lambda = \frac{\sqrt{\frac{\Gamma\underline{B}B_{opt}}{\chi}} - \underline{B}}{B_{opt} - \underline{B}}, \quad \text{and} \quad \delta_{opt} = \frac{B_{opt}}{1 - B_{opt}}\frac{(\tau_e + \tau)}{\tau_\eta}. \tag{18}$$

In the absence of strategic externalities ($r = 0$) and when there is no direct impact of aggregate volatility on payoffs ($u_{KK} = 0$), then $\chi = 0$. In this case, players choose to over-estimate the precision of the public signal as this reduces perceived fundamental uncertainty. As strategic considerations become more important (i.e., $|r|$ and hence χ increases), players have an incentive to believe that public information is noisier. On the other hand, if the cost of aggregate volatility increases (i.e., u_{KK} and hence χ becomes more negative), players have an incentive to believe the signal is more precise.

More generally, the above result mirrors Proposition 1, but adjusted for the fact that players now account for their experienced utility loss. When χ is sufficiently high, all players still choose to under-react to the public information by choosing $\delta_{\eta,i} = \underline{\delta}$. When

χ is sufficiently low, there also exists a unique symmetric equilibrium; however, when ψ is sufficiently high, this equilibrium choice is interior, i.e., the optimal $\delta_{sym} < \bar{\delta}$ due to the increased cost of deviating from rational expectations. Moreover, this interior δ_{sym} reflects the impact of aggregate errors on optimal beliefs; holding fixed ψ, δ_{sym} decreases in χ in this region, i.e., agents choose to believe the public signal is less informative as χ increases.

For intermediate χ, players endogenously disagree: players mix between δ_{opt} and $\underline{\delta}$, and the measure λ who choose δ_{opt} decreases in χ as before. Now, δ_{opt} (and the corresponding expressions for B_{opt} and λ) reflects the trade-off between desirable and accurate beliefs. When ψ is relatively low, the fraction λ of players over-react maximally (since $B_{opt} = \bar{B}$, we have $\delta_{\eta,i} = \delta_{opt} = \bar{\delta}$).[21] However, when experienced utility is more important (i.e., for larger ψ), players optimally choose an interior weight: $\delta_{opt} < \bar{\delta}$, reflecting the cost of holding extreme beliefs.

The region of endogenous disagreement now also depends on the relative importance of experienced utility, ψ. The following result establishes sufficient conditions for the existence of such an equilibrium.

Corollary 3. *If $\underline{\delta} \leq \frac{2(\tau_e+\tau)}{3(\tau_e+\tau)+\tau_\eta}$, then for any finite ψ, there exists a range of χ for which the unique equilibrium features endogenous disagreement.*

This result suggests that as long as individuals are permitted to underestimate the precision of the public signal sufficiently, there exist parameters that sustain a mixed equilibrium.[22] Note that for $\psi = 0$, $B_{opt} = \bar{B} > \underline{B}$, and that B_{opt} (weakly) decreases as ψ increases. If $\underline{\delta}$ (and consequently, $\underline{B}$) is sufficiently low, then we can show that $B_{opt} > \underline{B}$ for all finite ψ. Intuitively, in this case, there always exists some sufficiently high χ which ensures that choosing a sufficiently low $\underline{\delta}$ is optimal for some players, and this allows for the possibility of endogenous disagreement for any finite ψ.[23]

5 Application to Professional Forecasters

Our model can be applied to study the implications of wishful thinking to many settings, including those with investment complementarities, oligopolistic production, and informational spillovers in financial markets – we describe these in Appendix C. However, we

[21] Note that as $\psi \to 0$, $B_{opt} \to \bar{B}$ and $\Gamma \to 1$ so that the threshold χ and equilibrium fraction λ coincide with the analogous expressions in Proposition 1.

[22] At first glance, this result might seem at odds with the intuition that as $\psi \to \infty$, all players converge to exhibiting rational expectations. However, note that the convergence to rational expectations holds in the limit when we keep all other parameters fixed. In contrast, the current result shows that for any finite ψ arbitrarily large, there exists a region of the parameter space such that the unique equilibrium is mixed if $\underline{\delta}$ can be sufficiently low.

[23] This is straightforward to see since by taking $\psi \to \infty$, $B_{opt} \to \max\{B_{RE} - \underline{B}/2, \underline{B}\}$ and $\Gamma \neq 0$.

focus on an economically important and empirically relevant application — understanding the behavior of strategic professional forecasters.

One might be led to believe that professional forecasters - given their training, experience, and access to data - would be largely immune to biases such as wishful thinking. However, a substantial earlier literature emphasizes the importance of strategic considerations that affect the incentives of such forecasters to distort their reports: Some forecasters may have incentives to "stand out" from the rest, while others have incentives to shade their forecasts towards consensus in order to avoid being "too wrong".[24] These pressures can induce wishful thinking and subtly shift beliefs toward more favorable outcomes, especially when there is enough ambiguity in the data to allow for motivated interpretation without overtly violating statistical norms. In this sense, wishful thinking becomes not just a quirk, but a behavioral bias that is amplified by professional incentives.

Building on this insight, we consider a special case of our model to study strategic forecasts. Specifically, we assume that forecaster i's problem is to submit a forecast k_i of a target random variable θ that maximizes

$$u_i \equiv -(k_i - \theta)^2 - \rho (k_i - K)^2 - \gamma (K - \theta)^2 . \tag{19}$$

The first term captures the incentive to produce an accurate forecast. In the second term, the parameter $\rho \in (-1, 1)$ reflects the strategic considerations that the forecaster faces: when $\rho < 0$, forecasters want to stand out (i.e., forecasts are strategic substitutes), while when $\rho > 0$, forecasters want to herd (i.e., forecasts are strategic complements).[25]

The third term reflects the pure externality that aggregate forecasts have on an individual's payoffs. Importantly, when $\gamma < 0$, forecaster i is better off when the consensus forecast is further from the truth. This captures a natural feature of many forecasting settings that "being right" is more valuable when "others are wrong."

We argue that this type of externality is particularly relevant in financial market settings. Banerjee et al. (2024) provides a microfoundation for $\gamma < 0$ by showing that, in a trading setting with heterogenously informed investors, each investors' private information is more valuable when others are not informed and the price is noisy. Empirically, Bouchaud et al. (2019) show that when consensus analyst forecasts of earnings per share (EPS) deviate significantly from fundamentals, the market prices of such firms are also substantially misaligned relative to fundamentals, as evidenced by the pronounced profitability anomaly. To the extent that such anomalies represent profitable trading opportunities, an analyst

[24] See Croushore (1997), Ottaviani and Sørensen (2006), Marinovic, Ottaviani, and Sørensen (2010), Gemmi and Valchev (2023), and Marinovic, Ottaviani, and Sorensen (2013) for an important survey.

[25] Building on the insights of Morris and Shin (2002), Gemmi and Valchev (2023) show that when forecasts exhibit strategic substitutability (i.e., have an incentive to stand out from others), one can jointly explain the underreaction in consensus forecasts and overreaction in individual forecasts documented in the literature.

forecasting financial variables like EPS is better off when others' forecasts are more incorrect on average. This suggests that the externality among such analysts is characterized by a negative value of γ.[26]

Given her choice of subjective beliefs, forecaster i's optimal choice (which maximizes Equation (19)) is

$$k_i(s_i, s; \delta_{\eta,i}) = \frac{1}{1+\rho}\mathbb{E}_i\left[\theta + \rho K | s_i, s\right].$$

It is worth noting that the optimal forecast is a function of ρ, but independent of γ, so that with rational expectations, the externality that aggregate actions have on an individual's payoffs (γ) does not affect her forecast. In contrast, when forecasters choose their subjective beliefs, we show that both strategic considerations (i.e., ρ) and externalities (i.e., γ) have an impact on their subjective beliefs, and consequently, their forecasts. To see why, note that χ can be expressed as:[27]

$$\chi = -\rho - \gamma(1+\rho). \tag{20}$$

The expression clarifies that the relative impact of aggregate errors depends on both strategic considerations ρ and explicit externalities γ. When $\gamma = 0$, the above expression simplifies to $\chi = -\rho$: this implies that aggregate errors channel has a positive (negative) effect when forecasts are strategic substitutes (complements). On the other hand, when $\rho = 0$, so there are no strategic considerations, we have $\chi = -\gamma$. Intuitively, when a forecaster is better off when others are more wrong (i.e., the externality is negative $\gamma < 0$), the impact of aggregate errors is positive. Importantly, this highlights that wishful thinking about the precision of public information can affect subjective beliefs and behavior even in the absence of strategic considerations.

Given these observations, we can now apply Proposition 2 to get the following result.

Corollary 4. *Let* $\underline{\gamma} = -\frac{\frac{B_{opt}}{\underline{B}}\Gamma+\rho}{1+\rho}$ *and* $\bar{\gamma} = -\frac{\frac{\underline{B}}{B_{opt}}\Gamma+\rho}{1+\rho}$, *then*

1. *If* $\gamma < \underline{\gamma}$, *then a unique equilibrium exists, and is such that all players choose* $\delta_{\eta,i} = \underline{\delta}$.
2. *If* $\gamma > \bar{\gamma}$, *then a unique equilibrium exists, and is such that all players choose* $\delta_{\eta,i} = \delta_{sym}$ *(defined in equation (17)).*
3. *If* $\gamma \in [\underline{\gamma}, \bar{\gamma}]$, *then a unique equilibrium exists, and is such that a fraction* λ *of players choose* $\delta_{\eta,i} = \delta_{opt}$ *(defined in equation (18)), while others choose* $\delta_{\eta,i} = \underline{\delta}$.

[26] In Appendix B.2, we also allow forecasters' payoffs to depend on forecast dispersion. We show that this provides an incentive for individuals to distort their beliefs about the precision and correlation of others' signals; in what follows, we abstract from dispersion-based externalities for expositional clarity.

[27] This is because, for the payoff in (19), $u_{KK} = -2\rho - 2\gamma$, $u_{kk} = -2 - 2\rho$ and $u_{kK} = 2\rho$ and we can substitute these into expression (13).

While Morris and Shin (2002) (and more recently Gemmi and Valchev (2023)) clarify how strategic considerations can distort players' *actions* (strategic forecasts), our analysis shows that such considerations can endogenously distort their *subjective beliefs* too. Moreover, the above result suggests that holding fixed strategic considerations (i.e., ρ), the intensity of aggregate externalities (γ) can affect subjective beliefs. For instance, when γ is sufficiently large, there exists a unique symmetric equilibrium in which all forecasters believe that the public information is more precise than it actually is. In contrast, when γ is sufficiently negative so that one is better off when others make larger mistakes, there exists a unique symmetric equilibrium in which all forecasters believe that the public signal is noisier than it is. For intermediate values of γ, our model features endogenous disagreement about the interpretation of the public information.

Next, we characterize various implications of this result that help us understand the evidence from professional forecaster surveys.

5.1 Implication 1: Forecast error predictability

Coibion and Gorodnichenko (2012, 2015) develop a new approach for testing the departures of forecasts from rational expectations by regressing forecast errors on forecast revisions. Under the null of rational expectations and full information, the forecast errors are unpredictable, and so the regression coefficient is zero. However, they document that the regression coefficient for consensus forecasts is significantly positive in the data, suggesting that consensus forecasts under-react to available information. In contrast, Bordalo et al. (2020) documents that individual forecasts tend to exhibit overreaction: the analogous regression coefficients are negative. We characterize conditions under which these puzzling results can arise in our model, and provide novel testable predictions in these cases.

Given the optimal choice k_i of forecaster i, the forecast revision (FR) and forecast error (FE) are given by

$$FR_i(s, s_i) \equiv k_i(s, s_i) \quad \text{and} \quad FE_i \equiv \theta - k_i(s, s_i) \tag{21}$$

respectively. Specifically, the forecast revision FR_i reflects the update in forecaster i's forecast after she conditions on public and private signals, relative to her unconditional forecast before observing any information. Since we have assumed that all random variables have mean zero, this unconditional forecast is also zero. Taking averages across i gives us the analogous expressions for the consensus forecast revision $\bar{FR}$ and consensus forecast error $\bar{FE}$. One can then characterize the "Coibon-Gorodnichenko regression coefficient" for individual forecasts as CG_i and for consensus forecasts as CG_a, where

$$CG_i = \frac{\text{cov}(FE_i, FR_i)}{\text{var}(FR_i)}, \quad \text{and} \quad CG_a = \frac{\text{cov}(\bar{FE}, \bar{FR})}{\text{var}(\bar{FR})}. \tag{22}$$

We begin by characterizing conditions under which our model generates the empirically observed patterns for CG_a and CG_i in a symmetric equilibrium.

Proposition 3. ***Symmetric equilibrium.*** *Consider a symmetric equilibrium in which all forecasters choose $\delta_{\eta,i} = \delta^*$.*

(1) When forecasters exhibit rational expectations, then (i) individual forecasts exhibit overreaction ($CG_i < 0$) if and only if $\rho < 0$, and (ii) consensus forecasts exhibit underreaction ($CG_a > 0$) if $\tau_e > 0$.

(2) When forecasters exhibit wishful thinking then:

(i) In the absence of strategic considerations ($\rho = 0$) and explicit externalities ($\gamma = 0$), individual forecasts exhibit overreaction ($CG_i < 0$) and consensus forecasts exhibit underreaction ($CG_a > 0$) if τ_e is sufficiently large.

(ii) There is over-reaction at the individual level (i.e., $CG_i < 0$) iff $\tau_e\rho + \tau_\eta (1+\rho)^2 \delta^* (1-\delta^*) < 0$. This holds if γ is sufficiently large. For γ sufficiently small, $CG_i > 0$.

(iii) There is under-reaction at the consensus level (i.e., $CG_a > 0$) iff $\tau_e + \tau_\eta (1+\rho) \delta^* (1-\delta^*) > 0$. This holds if τ_e is sufficiently large.

Part (1) of the above result is analogous to the results of Gemmi and Valchev (2023). Under rational expectations ($\delta^* = 1$), forecasters condition correctly on all their information. In this case, individual forecasts exhibit overreaction or underreaction if and only if forecasters face strategic considerations. Individual forecasts tend to exhibit over-reaction when forecasters have an incentive to stand out (i.e., $\rho < 0$) but exhibit under-reaction when they have an incentive to herd (i.e., $\rho > 0$). Moreover, consensus forecasts can exhibit underreaction even in the absence of strategic considerations (i.e., if $\rho = 0$): since individuals have dispersed private signals with uncorrelated errors, the aggregate information available to them is more precise than any individual's information, and so the consensus forecast under-reacts to the aggregate information.[28]

Part (2) of the above result highlights the novel mechanism in our setting. Importantly, when forecasters exhibit wishful thinking, individual forecasts exhibit overreaction and consensus forecasts can exhibit underreaction even in the absence of strategic considerations ($\rho = 0$) and explicit externalities ($\gamma = 0$). Intuitively, this is because in the absence of such considerations, a forecaster's objective is to choose an action as close to the target as possible,

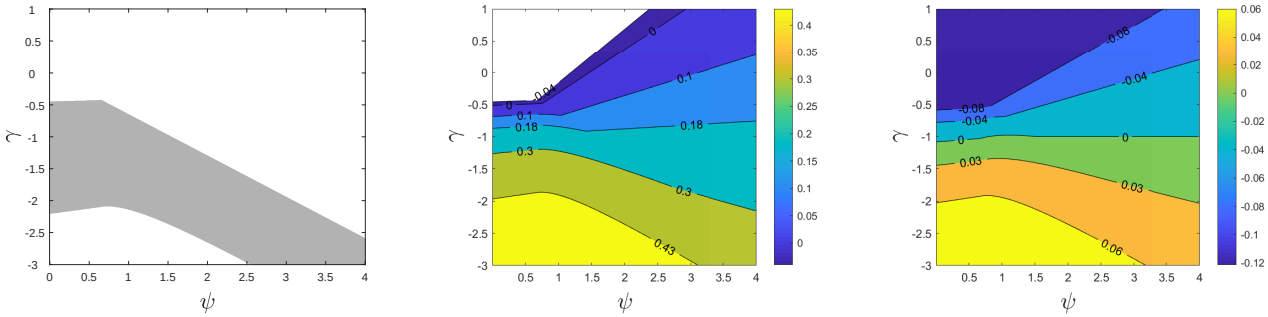
[28]This is consistent, for instance, with the noisy information model of Woodford (2001), which Coibion and Gorodnichenko (2015) propose as an explanation for their evidence on aggregate under-reaction.

and so her anticipatory utility is higher when she perceives the public signal to be more informative. This naturally leads to overreaction at the individual level. The underreaction at the consensus level occurs when private information is sufficiently informative because of aggregation, as in the case of rational expectations.

More generally, individuals who exhibit wishful thinking have an incentive to distort their subjective beliefs about the public information depending on strategic considerations (i.e., ρ) and the externality induced by the aggregate forecast (i.e., γ). In particular, for a given level of ρ, this leads to an over-weighting of public information when γ is sufficiently high and an under-weighting of public information when γ is sufficiently low. Figure 2 provides a numerical illustration of these results for $\rho = 0$, and establishes that similar results extend to the mixed equilibrium: for a given ψ, both individual and consensus regression coefficients increase as γ decreases.

Figure 2: Average CG_a and CG_i

The figure plots the CG_a and CG_i as a function of ψ and γ. Panel (a) characterizes the region of mixed equilibrium in gray, panel (b) shows a contour plot of the average consensus CG coefficient CG_a, and panel (c) shows a contour plot of the average individual CG coefficient CG_i. Other parameters: $\tau = \tau_e = \tau_\eta = 1$, $\rho = 0$ and $\underline{\delta} = 1/\bar{\delta} = 0.55$.



(a) Region of disagreement (b) CG_a contour plot (c) CG_i contour plot

This mechanism also provides a prediction of our model that distinguishes it from other explanations proposed in the literature. Importantly, a stylized fact from the empirical evidence is that while consensus forecasts exhibit underreaction across most variables, individual forecasts tend to exhibit over-reaction for macroeconomic targets, but under-reaction for financial forecasts. For instance, Bordalo et al. (2020) and Gemmi and Valchev (2023) show that individual forecasts of short-term treasuries exhibit under-reaction, while Bouchaud et al. (2019) shows that analyst-level forecasts for earnings per share (EPS) exhibit under-reaction. Existing models often target either individual-level over-reaction or under-reaction, and therefore cannot usually reconcile both in the same framework.[29]

[29]For instance, Bordalo, Gennaioli, Porta, and Shleifer (2019), Bordalo et al. (2020), Afrouzi et al. (2023),

In contrast, our analysis clarifies how a single mechanism can generate both sets of results depending on the nature and magnitude of the underlying payoff externalities. Specifically, for relatively larger (or less negative) values of the externality parameter γ, individual forecasts exhibit overreaction ($CG_i < 0$), while the consensus forecast exhibits underreaction ($CG_a > 0$). This region corresponds to the empirical behavior of macroeconomic forecasts. For sufficiently negative values of γ, individual forecasts exhibit underreaction ($CG_i > 0$), while the consensus forecast continues to exhibit underreaction ($CG_a > 0$). This region corresponds to the empirical behavior of financial forecasts, such as earnings or interest-rate forecasts, where one might expect externalities to be negative. Intuitively, in these cases, a larger error in the aggregate/consensus forecasts suggests a potentially profitable trading opportunity and consequently leads to higher payoffs.[30]

The following result is another distinctive prediction of our model.

Proposition 4. ***Dispersion in*** CG ***coefficients.*** *In a symmetric equilibrium, the dispersion of CG_i coefficient is zero. In a mixed equilibrium, the dispersion of CG_i coefficients is positive and hump-shaped in γ.*

In a mixed equilibrium, the individual CG coefficients are different across investors. This prediction is consistent with the heterogeneity in individual CG coefficients that Bouchaud et al. (2019) document for EPS forecasts and the disagreement about public information that Ahn and Farmer (2024) document for macroeconomic forecasts. Moreover, to the extent that we expect γ to be large and negative for financial market variables but close to zero for macroeconomic variables, we expect that dispersion in individual CG coefficients for financial market forecasts is greater than or equal to the dispersion for macroeconomic forecasts.[31]

Existing models with exogenously assumed biases often assume that individuals are symmetric, and so do not give rise to heterogeneity in individual CG forecasts. Moreover, in models with rational inattention or endogenous information acquisition, this heterogeneity does not arise if individuals have rational expectations, even if they have different precisions of private information. Recall that dispersion in beliefs (and actions) can arise either because players have private signals or because they disagree about the interpretation of the public signal, and so dispersion in forecasts cannot be used to distinguish these settings. However, even when private signal precisions differ across players (e.g., due to different information

and Bordalo, Gennaioli, Porta, and Shleifer (2024) generate over-reaction at the individual level, while Bouchaud et al. (2019) generates under-reaction.

[30] As discussed above, this intuition is formalized in Banerjee et al. (2024), who show that, in a trading setting, anticipatory utility is increasing in the noisiness of the price, all else equal.

[31] Note that Corollary 4 implies that for γ close to zero or positive, we have a symmetric equilibrium.

acquisition decisions, or staggered attention), there cannot be a dispersion in CG_i coefficients when players exhibit rational expectations, since they use all their information efficiently (and so $CG_i = 0$).

Such dispersion arises in our model when players endogenously choose to disagree about public information. Moreover, in contrast to models that exogenously assume that players agree to disagree about public signals, our model predicts that this dispersion is more likely to arise when public information precision is lower (see Corollary 2). There is recent empirical evidence (e.g., Ahn and Farmer (2024)) that suggests disagreement about the interpretation of public information increases during economic downturns. To the extent that such periods are also associated with high uncertainty driven by a decrease in the precision of public information, this evidence is consistent with our model's predictions.

5.2 Implication 2: Aggregate response to public information

Based on SPF forecasts of output growth and inflation, Kohlhas and Walther (2021) show that individual forecasts exhibit underreaction to new information in addition to overreaction to recent realizations. In our context, the empirical specifications that they consider map into the following:[32]

$$FE_i = a_{s,i} + b_{s,i}s + e_{s,i}, \quad \text{and} \quad FE_i = a_{f,i} + b_{f,i}\bar{FR} + e_{f,i} \tag{23}$$

The full information, rational expectations benchmark implies that both $b_{s,i}$ and $b_{f,i}$ are zero. However, Kohlhas and Walther (2021) document that $b_{s,i} < 0$, which is indicative of overreaction to recent realizations, and $b_{f,i} > 0$, which implies underreaction to new information.

The following result characterizes conditions under which the above hold in our setting.

Proposition 5. *Consider a symmetric equilibrium in which all forecasters choose $\delta_{\eta,i} = \delta^*$.*

(i) There is overreaction to public information (i.e., $b_{s,i} < 0$) if and only if $\delta^* > 1$. This holds if and only if $\gamma > -1$.

(ii) There is underreaction to consensus forecast revisions (i.e., $b_{f,i} > 0$) if and only if $\tau_e + (1+\rho)\,\tau_\eta \delta^* (1-\delta^*) > 0$.

Both conditions hold in the absence of strategic considerations ($\rho = 0$) and explicit externalities ($\gamma = 0$) if τ_e is sufficiently large.

The intuition for these results is similar to that of the earlier results. When the externality due to aggregate forecasts is not too large (i.e., $\gamma > -1$), forecasters have an incentive to

[32] In Kohlhas and Walther (2021), the first specification regresses forecast errors on lagged realizations of the target variable. Since our model is static, we replace this with a regression on the public signal s.

believe that the public information is more informative than it actually is. This implies individuals tend to overreact to public information. On the other hand, when the private signals are sufficiently precise, aggregation implies that the consensus forecast revision is more informative than individuals perceive it to be. As a result, individual forecasts appear to underreact to consensus forecast revisions.

Importantly, our model is able to match these results even in the absence of strategic considerations or externalities, which distinguishes us from existing approaches. Specifically, Kohlhas and Walther (2021) argues that their evidence is inconsistent with existing models of expectations formation: noisy rational expectations models cannot reconcile the evidence on overreaction, while behavioral approaches (e.g., overconfidence, extrapolation, diagnostic expectations) cannot reconcile the underreaction to new information. Moreover, unlike the baseline specification of Kohlhas and Walther (2021), our model is also able to generate overreaction in individual forecasts as measured by CG_i (as documented by Bordalo et al. (2020)).[33]

5.3 Implication 3: Belief dispersion and public information

There is a large literature in macroeconomics and finance that documents how belief dispersion among investors and professional forecasters varies predictably with the business cycle (e.g., Mankiw, Reis, and Wolfers (2003), Andrade, Crump, Eusepi, and Moench (2016)), and can increase after public announcements (e.g., Kandel and Pearson (1995)). More recently, Ahn and Farmer (2024) decomposes the belief dispersion across professional forecasters into differences in individuals' long-term beliefs (or priors), differences in private information, and differences in the interpretation of public information. They show that while belief dispersion is primarily driven by private information in normal times, heterogeneous reactions to public information become a key driver of disagreement during economic downturns.

Our analysis provides a mechanism for why such disagreement can arise *endogenously*, and so sheds light on "what happens when." Given optimal forecasts k_i, we can define the cross-sectional dispersion in forecasts as

$$\sigma_k^2 = \mathbb{E}\left[\mathrm{var}[k_i|\{\theta,\eta\}]\right], \tag{24}$$

which reflects the expected cross-sectional variance in forecasts across players given fundamentals and public noise. Let $\sigma_{k,RE}^2$ denote the corresponding measure if forecasters exhibit rational expectations. The following result characterizes how forecast dispersion

[33] Kohlhas and Walther (2021) claim that the evidence of overreaction in individual forecasts is driven by outliers and provide a generalized model with a bias that is able to generalize this result. Our model naturally gives rise to both results in the same setting due to a transparent and intuitive mechanism.

varies with strategic considerations, payoff externalities and public information quality.

Proposition 6. *(i) If players' exhibit rational expectations, then dispersion in forecasts decreases with public information quality i.e.,* $\frac{\partial \sigma^2_{k,RE}}{\partial \tau_\eta} < 0$*, and is independent of* γ*.*

(ii) In a symmetric equilibrium with subjective beliefs, dispersion in forecasts is higher than under rational expectations (i.e., $\sigma^2_k > \sigma^2_{k,RE}$*) if and only if* $\gamma < -1$*. Moreover, dispersion decreases with public information quality i.e.,* $\frac{\partial \sigma^2_k}{\partial \tau_\eta} < 0$ *and decreases with* γ*.*

(iii) In a mixed equilibrium with subjective beliefs, dispersion in forecasts arises because of different weights on public and private information. It can increase with public information quality i.e., $\frac{\partial \sigma^2_k}{\partial \tau_\eta} > 0$*, and is non-monotonic in* γ*.*

(iv) For sufficiently low ψ*, all else equal, the likelihood of a mixed equilibrium (with heterogeneous interpretation of public information) is higher when the quality of public information is low (i.e.,* τ_η *is low).*

Part (i) provides the rational expectations benchmark when all forecasters have correct beliefs about the public signal. As a result, all else fixed, when public information is more precise, forecasters rationally put less weight on their private signals, and so forecast dispersion falls. Moreover, since forecasters' actions only depend on the informational content of their signals and on strategic considerations (i.e., ρ), forecast dispersion is independent of the aggregate externality parameter γ.

Part (ii) characterizes forecast dispersion in the symmetric equilibrium. When the relative impact of aggregate errors increases, forecasters endogenously choose to put less weight on the public signal (i.e., δ_{sym} is decreasing in χ), which in turn, increases dispersion in forecasts. As a result, dispersion is higher in the symmetric equilibrium than under rational expectations when χ is sufficiently large (i.e., $\chi > 1$). On the other hand, because forecasters agree on their interpretation of the public signal, we show that forecast dispersion decreases in τ_η. Notably, this corresponds to the evidence on forecast dispersion in "normal times" as documented by Ahn and Farmer (2024).

In contrast, part (iii) describes how dispersion behaves when players endogenously *disagree* about the precision of the public signal. In this case, forecast dispersion is driven by both (i) dispersion in private information and (ii) heterogeneity in the interpretation of the public signal, and corresponds to Ahn and Farmer (2024)'s evidence on forecast dispersion during economic downturns. Recall from Proposition 2 that λ, the fraction of players who maximally over-react to the public signal, is monotonic in χ. As a result, disagreement about the public signal and dispersion in forecasts, is non-monotonic in χ as the measure of

each type shifts endogenously.[34] Moreover, an increase in τ_η can increase forecast dispersion *across* the two groups: the distance in conditional expectations between those who choose $\underline{\delta}$ and those who choose δ_{opt} can increase when the precision of the public signal improves.

Recent studies have found that increased communication by central banks can increase forecast dispersion (e.g., Lustenberger and Rossi (2018)), consistent with this prediction. Such evidence is difficult to reconcile in models with symmetrically informed individuals who exhibit rational expectations. In such settings, providing a more precise public signal usually crowds out the acquisition of private information and consequently reduces dispersion in beliefs. In principle, one may be able to generate higher belief dispersion in response to more public information with (i) endogenous information acquisition, (ii) heterogeneous inattention, or (iii) uncertainty about the precision of the public information, where individuals learn privately about this precision. However, to the extent that individuals have correct priors and update their beliefs using Bayes' rule (i.e., exhibit rational expectations), we expect that individual forecast errors are orthogonal to forecast revisions, i.e., individual CG coefficients are zero. In contrast, as we clarify in Section 5.1, our model predicts that individual CG coefficients are negative even in the absence of strategic considerations and externalities and heterogeneous in the mixed equilibrium.

Finally, part (iv) of the above result implies that all else equal, the mixed equilibrium in which forecasters disagree about the interpretation of public information is more likely to arise when public information quality is low. To the extent that public information quality is worse and public uncertainty is higher during economic downturns, this prediction is broadly consistent with Ahn and Farmer (2024) evidence about when disagreement about public information drives forecast dispersion.

5.4 Comparison to existing theories

In this section, we summarize a list of stylized facts about professional forecasts and discuss how these allow us to distinguish our framework from existing theories of expectation formation. Although our goal is to provide distinctive predictions, we view our mechanism as complementary to the alternatives we discuss below, and expect that a number of them are relevant in practice.

Coibion and Gorodnichenko (2012, 2015) focused on documenting underreaction to forecast revisions for consensus forecasts (i.e., $CG_a > 0$). Since then, the literature has documented a number of stylized facts about individual forecasts:

1. (i) Overreaction to forecast revisions for macroeconomic variables, i.e., $CG_i < 0$, but (ii) underreaction for financial variables i.e., $CG_i > 0$ (e.g., Bordalo et al. (2020) and

[34] Disagreement is low when λ is close to zero or one, but higher in the middle.

Bouchaud et al. (2019)).

2. Cross-sectional heterogeneity in individual CG coefficients (e.g., Bouchaud et al. (2019)).
3. Overreaction to public information (e.g., Kohlhas and Walther (2021)).
4. Underreaction to consensus forecast revision (e.g., Kohlhas and Walther (2021)).
5. Disagreement in the interpretation of public information (e.g., Kandel and Pearson (1995), Ahn and Farmer (2024)), which increases during economic downturns.

Note that the results from the earlier sections imply that our model's predictions are consistent with all the above stylized facts. In fact, our model gives rise to consensus underreaction and (1)(i), (3) and (4) even in the absence of strategic considerations and externalities (i.e., when $\rho = \gamma = 0$), and naturally gives rise to 1(ii), (2) and (5) when γ is negative but not too small (and $\rho = 0$). In contrast, existing models are unable to generate some of the above predictions.

Rational inattention. Traditional models of noisy rational expectations or rational inattention (e.g. Woodford (2001), Maćkowiak and Wiederholt (2009), Myatt and Wallace (2012), Kohlhas and Walther (2021)) can generate consensus underreaction and (4). However, such models (i) focus on symmetric equilibria and (ii) assume players have correct beliefs about the information they observe and update their beliefs using Bayes' rule. The former implies that such models are unlikely to be able to generate (2) or (5). The latter usually implies that CG coefficients are zero, and so do not naturally give rise to (1), (2) or (3), even if they pay attention at different times or have private information of different precisions.[35]

Strategic forecasters. While traditional models of forecasters that incorporate strategic considerations (e.g., Gemmi and Valchev (2023)) can generate both consensus underreaction and (1), they often assume that investors are symmetric and so cannot generate (2) or (5).

Behavioral models. Recent models of expectation formation have focused on jointly explaining consensus underreaction and individual overreaction. These include models of diagnostic expectations (e.g., Bordalo et al. (2020)), extrapolation (e.g., Barberis et al. (2015)), overconfidence (e.g., Daniel et al. (1998)), optimally noisy memory (e.g., Da Silveira, Sung, and Woodford (2020), Sung (2024)), and costly information retrieval (e.g., Afrouzi et al. (2023)). However, the benchmark versions of these models assume symmetric

[35]Recently, Kohlhas and Walther (2021) shows that forecasters with rational but limited and asymmetric attention - allocating more attention to volatile, procyclical components of the economy - can simultaneously exhibit extrapolation from recent trends and underreaction to new information, i.e., explain (3) and (4). However, their baseline model cannot explain (2) and (5), which rely on heterogeneous reactions to public information.

forecasters and so are generally unable to generate (2) and (5), and given their focus, generally do not generate underreaction in individual forecasts i.e., part (ii) of (1), or underreaction to new information, i.e., (4).

A related class of papers explain under- and over-reaction through selective memory rather than distorted signal processing (e.g., d'Arienzo, 2020; Bordalo, Conlon, Gennaioli, Kwon, and Shleifer, 2023; and Gennaioli, Leva, Schoenle, and Shleifer, 2024). A key insight of this literature is that under- and over-reaction are two sides of the same mechanism: when agents recall a narrow set of extreme experiences, they overreact; when they neglect relevant past data, they underreact. In principle, such mechanisms can also generate endogenous disagreement about public information because agents with different experiences or ages retrieve different sets of memories when faced with the same information. A distinctive prediction of our mechanism is that belief distortions that generate the above prediction are driven by variation in strategic considerations and payoff externalities, and arise even when agents have perfect recall or share common experiences.

Difference of opinions models. In principle, models in which investors are assumed to interpret public information differently (e.g., Kandel and Pearson (1995), Scheinkman and Xiong (2003), Banerjee and Kremer (2010)) are consistent with (2), (3), (4) and (5). However, in contrast to such models, our mechanism provides a set of predictions that clarify when such disagreement is more likely to arise and when we are more likely to observe symmetric behavior. In particular, our model predicts it is more likely that we observe heterogeneous interpretations of public information when public information is not very informative and aggregate uncertainty is high.

6 Subjective beliefs about private information

Subjective beliefs about one's own private information. In Appendix B.1, we allow players to choose subjective beliefs about the quality of their own private information. Specifically, player i's subjective belief about the error in her private signal is given by

$$\varepsilon_i \sim_i N\left(0, \frac{1}{\delta_{e,i}\tau_e}\right), \tag{25}$$

where $\delta_{e,i} \in \left[\underline{\delta}, \overline{\delta}\right]$. In Proposition 7, we show that when players choose $\delta_{e,i}$ (only) to maximize their total utility, as in (6), there exists a unique, symmetric equilibrium. All players choose to exhibit over-confidence in their private information i.e., set $\delta_{e,i} > 1$ since beliefs about the quality of one's private signal affect only the fundamental uncertainty channel.[36] In Proposition 8, we consider a setting in which players choose their subjective beliefs about

[36] While managerial over-confidence has been extensively documented, our model predicts that over-confidence endogenously increases in prior uncertainty (i.e., decreases in τ), consistent with the empirical evidence on CFO beliefs found in Ben-David, Graham, and Harvey (2013).

both private and public information. In particular, we are able to fully characterize the equilibrium for all χ when the relative importance of experienced utility (ψ) is sufficiently low. The equilibria which arise are analogous to our benchmark. In particular, for intermediate χ, there exists a unique mixed strategy equilibrium that features endogenous disagreement about public information.

Subjective beliefs about the private information of others. In Appendix B.2, we allow players to choose their beliefs about others' private signals. Specifically, we assume that player i believes that the error in player j's private signal is given by

$$\varepsilon_j =_i \sqrt{1-p_i^2}\eta + p_i\varepsilon_j, \text{ where } \quad \eta, \varepsilon_j \sim N\left(0, \tfrac{1}{\delta_{j,i}\tau_e}\right), \tag{26}$$

η and ε_j are independent of each other and all other variables and $p_i \in [0,1]$ and $\delta_{j,i} \in \left[\underline{\delta}, \overline{\delta}\right]$ are chosen by player i to maximize her anticipatory utility. Notably, player i has subjective beliefs about the precision of others' signals ($\delta_{j,i}$) and the correlation (or commonality) across them (p_i). To highlight the impact of beliefs about others, we shut down the public signal, s, and assume players hold objective beliefs about the precision of their own signal.

In Proposition 9, we show that subjective belief choices depend on the relative importance of dispersion versus aggregate volatility. When higher dispersion increases payoffs by more than aggregate volatility (i.e., $u_{\sigma\sigma} > u_{KK}$), player i benefits from believing that others' signals are more independent (i.e., distorting p_i upwards). Moreover, if $u_{\sigma\sigma}$ is positive, then she wants to believe that $\delta_{j,i}$ is lower, since this further increases the dispersion in players' actions; otherwise she chooses to increase $\delta_{j,i}$. On the other hand, when aggregate volatility increases payoffs by more ($u_{\sigma\sigma} < u_{KK}$), player i prefers to distort p_i downwards, and if $u_{KK} > 0$, player i lowers $\delta_{j,i}$. These distortions serve to increase the perceived volatility of aggregate actions.

A player's action (i.e., k_i) does not depend on her subjective beliefs about others' private signals, because the aggregate action is invariant in the dispersion of individual signals and depends only on θ and η. In a more general setting, where players also hold subjective beliefs about their own signal (or the public signal), beliefs about others' signals may have an indirect effect.[37] While such interactions are potentially interesting, we expect that the main economic channels are the same as in our main analysis.

7 Conclusions

In a standard model where players are uncertain about the precision of public information, we show that "wishful thinking" can lead to variation in, and endogenous disagreement

[37] For example, if player i believes that $p_i = 0$, she anticipates higher volatility of aggregate errors which will change the marginal benefit of believing the public signal is noisier.

about, the interpretation of public information. We apply this insight to the strategic behavior of forecasters and derive testable predictions about the aggregate response to public information, dispersion in beliefs, and implications for individual-level forecast error regressions that distinguish our model from the existing literature. We explore additional applications of our framework in Appendix C.

There are natural opportunities for future work. First, it would be interesting to study how information acquisition or attention interacts with players' endogenous perception in the presence of externalities. Second, one could extend the analysis to dynamic settings and characterize the implications of wishful thinking for the term structure of forecasts and disagreement, and how this interacts with the persistence of the fundamentals. Finally, recognizing that individuals are likely to exhibit wishful thinking in some settings but a preference for robust control under others, it would be interesting to explore the implications of a setting in which individuals can choose to exhibit both.

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A Proofs

A.1 Proof of Lemma 1

We follow the proof of Proposition 1 in Angeletos and Pavan (2007). Suppose player i conjectures that the average action has the form $K = \kappa_0 + \alpha\theta + \beta s$, where α and β are determined in equilibrium. Then we solve for the equilibrium by (i) plugging in the conjecture in the optimal action for player i, (ii) aggregating across players, and (iii) matching terms. This yields:

$$\alpha = \tfrac{A(1-r)\kappa_1}{1-rA}, \quad \beta = \tfrac{B}{1-rA}\kappa_1. \tag{27}$$

where $A \equiv \int_i A_i di$ and $B \equiv \int_i B_i di$ reflect the average weights players put on their private and public signals, respectively.[38] Note that $A, B < 1$ and $r < 1$ (by Assumption 1) ensure that the above solutions are well-defined and characterize a unique linear equilibrium.

A.2 Proof of Lemma 2

Let $\mathcal{K} \equiv rK + (1-r)\kappa$ denote the target action for each player. Given the assumptions about payoffs, note that we can express the utility as:

$$u(k, K, \theta, \sigma) = \begin{array}{c} u_0 + u_k k + u_K K + u_\theta \theta + \frac{1}{2}\left(u_{kk}k^2 + u_{KK}K^2 + u_{\theta\theta}\theta^2 + u_{\sigma\sigma}\sigma^2\right) \\ +u_{k\theta}k\theta + u_{K\theta}K\theta + u_{kK}kK \end{array}$$

A Taylor expansion around the target $(\kappa, \kappa, \theta, 0)$ yields:

$$u(k, K, \theta, \sigma) = \begin{array}{c} u(\kappa, \kappa, \theta, 0) + \frac{\partial}{\partial k}u(\kappa, \kappa, \theta, 0)\cdot(k-\kappa) + \frac{\partial}{\partial K}u(\kappa, \kappa, \theta, 0)\cdot(K-\kappa) \\ +\frac{1}{2}\left(u_{\sigma\sigma}\sigma^2 + u_{kk}(k-\kappa)^2 + u_{KK}(K-\kappa)^2 + 2u_{kK}(k-\kappa)(K-\kappa)\right) \end{array} \tag{28}$$

Note that $\frac{\partial}{\partial k}u(\kappa, \kappa, \theta, 0) = u_k + u_{kk}\kappa + u_{k\theta}\theta + u_{kK}\kappa = 0$ and

$$(k-\kappa)^2 = (k-\mathcal{K})^2 + r^2(K-\kappa)^2 + 2r(k-\mathcal{K})(K-\kappa)$$

Let $\mathbb{E}_j[\cdot]$ denote expectations w.r.t. arbitrary beliefs - we will later plug in subjective and objective beliefs. Then,

$$\mathbb{E}_j[u(k, K, \theta, \sigma)] = \mathbb{E}_j\left[\begin{array}{c} u(\kappa, \kappa, \theta, 0) + (u_K + ru_{kk}(k-\mathcal{K}) + u_{kK}(k-\kappa))\cdot(K-\kappa) \\ +\frac{1}{2}\left(u_{\sigma\sigma}\sigma^2 + u_{kk}(k-\mathcal{K})^2 + (u_{KK} + r^2u_{kk})(K-\kappa)^2\right) \end{array}\right] \tag{29}$$

Since $ru_{kk} = -u_{kK}$,we have that

$$ru_{kk}(k-\mathcal{K}) + u_{kK}(k-\kappa) = u_{kK}(\mathcal{K}-\kappa) = -r^2u_{kk}(K-\kappa)$$

which implies

$$\mathbb{E}_j[u(k, K, \theta, \sigma)] = \mathbb{E}_j\left[\begin{array}{c} u(\kappa, \kappa, \theta, 0) + u_K(K-\kappa) \\ +\frac{1}{2}\left(u_{\sigma\sigma}\sigma^2 + u_{kk}(k-\mathcal{K})^2 + (u_{KK} - r^2u_{kk})(K-\kappa)^2\right) \end{array}\right] \tag{30}$$

Player i's subjective beliefs about the joint distribution of $(\theta, s_i, \{s_j\}_{j\neq i}, s)$ are given by

$$\begin{pmatrix} \theta \\ s_i \\ s_j \\ s \end{pmatrix} \sim_i N\left(\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \frac{1}{\tau} & \frac{1}{\tau} & \frac{1}{\tau} & \frac{1}{\tau} \\ \frac{1}{\tau} & \frac{1}{\tau} + \frac{1}{\tau_e} & \frac{1}{\tau} & \frac{1}{\tau} \\ \frac{1}{\tau} & \frac{1}{\tau} & \frac{1}{\tau} + \frac{1}{\tau_e} & \frac{1}{\tau} \\ \frac{1}{\tau} & \frac{1}{\tau} & \frac{1}{\tau} & \frac{1}{\tau} + \frac{1}{\delta_{\eta,i}\tau_\eta} \end{pmatrix}\right). \tag{31}$$

[38] Implicitly, we are assuming that the law of large numbers implies: $\int_i A_i s_i di = \int_i A_i di \times \int_i s_i di = A \times \theta$. Specifically, this assumes there is no cross-sectional correlation between A_i and s_i, but this is valid because $(\delta_{e,i}, \delta_{\eta,i})$ are chosen before s_i (and s) are observed.

Let $u_0 \equiv u(\kappa, \kappa, \theta, 0)$ and note that the unconditional expectations $\mathbb{E}_i[u_0] = \mathbb{E}[u_0]$ and $\mathbb{E}_i[u_K(K-\kappa)] = \mathbb{E}[u_K(K-\kappa)]$ do not depend on player i's subjective beliefs. Then, player i's anticipatory utility is given by:

$$AU_i = \mathbb{E}_i[u_i] = \begin{array}{c} \mathbb{E}[u_0] + \mathbb{E}[u_K(K-\kappa)] + \frac{u_{\sigma\sigma}}{2}\sigma_i^2 \\ +\frac{1}{2}\left(u_{kk}\mathbb{E}_i\left[(k-\mathcal{K})^2\right] + (u_{KK} - r^2 u_{kk})\,\mathbb{E}_i\left[(K-\kappa)^2\right]\right) \end{array} \tag{32}$$

Moreover,

$$\mathbb{E}_i\left[(k-\mathcal{K})^2\right] = \left(\frac{\kappa_1(1-r)}{1-rA}\right)^2 \mathrm{var}_i\left[\theta|s_i, s\right]$$

$$\mathbb{E}_i\left[(K-\kappa)^2\right] = \left(\frac{\kappa_1(1-A-B)}{1-rA}\right)^2 \mathrm{var}\left[\theta\right] + \left(\frac{\kappa_1 B}{1-rA}\right)^2 \mathrm{var}_i\left[s|\theta\right]$$

Substituting all the above expressions into equation (32) and simplifying gives us

$$AU_i(\delta_{\eta,i}) = \bar{\mathcal{A}}_i + \left(\frac{\kappa_1}{1-rA}\right)^2 \left[u_{kk}(1-r)^2 \mathrm{var}_i\left[\theta|s_i, s\right] + \left(u_{KK} - r^2 u_{kk}\right) B^2 \mathrm{var}_i\left[s|\theta\right]\right], \tag{33}$$

where $\bar{\mathcal{A}}_i$ does not depend upon player i's choice of beliefs.

A.3 Proof of Corollary 1

Since all other players exhibit rational expectations ($\delta_{\eta,j} = 1 \forall j \neq i$), $B = \int_i B_i di = B_{RE}$, the objective of player i (from (6) and equation (33)) is

$$\max_{\delta_{\eta,i} \in \{\underline{\delta}, \bar{\delta}\}} \frac{B_{RE}^2 \chi}{\delta_{\eta,\mathrm{i}} \tau_\eta} - \frac{1}{\delta_{\eta,\mathrm{i}} \tau_\eta + \tau_e + \tau}$$

where player i takes $-u_{kk}\left(\frac{\kappa_1(1-r)}{1-rA}\right)^2 > 0$ as given. It is easy to see that the above objective is either downward sloping or U shaped in $\delta_{\eta,i}$ which implies that player i will choose either $\delta_{\eta,i} = \bar{\delta}$ or $\delta_{\eta,i} = \underline{\delta}$. Player i will choose $\delta_{\eta,i} = \bar{\delta}$ iff

$$\frac{B_{RE}^2 \chi}{\bar{\delta}\tau_\eta} - \frac{1}{\bar{\delta}\tau_\eta + \tau_e + \tau} > \frac{B_{RE}^2 \chi}{\underline{\delta}\tau_\eta} - \frac{1}{\underline{\delta}\tau_\eta + \tau_e + \tau}$$

which simplifies to the condition in Corollary 1.

A.4 Proof of Proposition 1

Since $\psi = 0$, the objective of player i (from (6) and equation (33)) can be rewritten as

$$\max_{\delta_{\eta,i} \in \{\underline{\delta}, \bar{\delta}\}} \frac{B^2 \chi}{\delta_{\eta,\mathrm{i}} \tau_\eta} - \frac{1}{\delta_{\eta,\mathrm{i}} \tau_\eta + \tau_e + \tau},$$

since player i takes $-u_{kk}\left(\frac{\kappa_1(1-r)}{1-rA}\right)^2 > 0$ as given. It is easy to see that the above objective is either downward/upward sloping or U shaped in $\delta_{\eta,i}$ which implies that player i will choose either $\delta_{\eta,i} = \bar{\delta}$ or $\delta_{\eta,i} = \underline{\delta}$. Conjecture that all other players choose $\delta_{\eta,j} = \bar{\delta} \forall j \neq i$. In this case, $B = \bar{B}$ and player i will also choose $\delta_{\eta,i} = \bar{\delta}$ iff

$$\frac{\bar{B}^2 \chi}{\bar{\delta}\tau_\eta} - \frac{1}{\bar{\delta}\tau_\eta + \tau_e + \tau} > \frac{\bar{B}^2 \chi}{\underline{\delta}\tau_\eta} - \frac{1}{\underline{\delta}\tau_\eta + \tau_e + \tau}$$

which is true iff $\chi < \underline{B}/\bar{B}$. Similarly, conjecture that all players choose $\delta_{\eta,i} = \underline{\delta}$. Player i will also choose $\delta_{\eta,i} = \underline{\delta}$ iff

$$\frac{\underline{B}^2\chi}{\bar{\delta}\tau_\eta} - \frac{1}{\bar{\delta}\tau_\eta + \tau_e + \tau} < \frac{\underline{B}^2\chi}{\underline{\delta}\tau_\eta} - \frac{1}{\underline{\delta}\tau_\eta + \tau_e + \tau}$$

which is true iff $\chi > \bar{B}/\underline{B}$. This also implies that, if $\chi \in \left[\underline{B}/\bar{B}, \bar{B}/\underline{B}\right]$, then a symmetric equilibrium cannot exist. Conjecture an equilibrium in which a fraction λ of players choose $\delta_{\eta,j} = \bar{\delta}$ and the remaining choose $\delta_{\eta,j} = \underline{\delta}$. This will be the equilibrium iff player i also is indifferent between these choices i.e.,

$$\frac{B^2\chi}{\bar{\delta}\tau_\eta} - \frac{1}{\bar{\delta}\tau_\eta + \tau_e + \tau} = \frac{B^2\chi}{\underline{\delta}\tau_\eta} - \frac{1}{\underline{\delta}\tau_\eta + \tau_e + \tau}$$

which will be true iff $B = \sqrt{\frac{\bar{B}\underline{B}}{\chi}}$. By definition, $B = \lambda\bar{B} + (1-\lambda)\underline{B}$, which implies $\lambda = \frac{\sqrt{\frac{\bar{B}\underline{B}}{\chi}} - \underline{B}}{\bar{B} - \underline{B}}$. Finally, note that $\lambda \in (0,1) \iff \chi \in \left[\frac{\underline{B}}{\bar{B}}, \frac{\bar{B}}{\underline{B}}\right]$. This implies that the equilibrium is continuous.

A.5 Proof of Corollary 2

Define

$$\bar{\chi} \equiv \frac{\bar{B}}{\underline{B}} = \frac{\bar{\delta}\left(\underline{\delta}\tau_\eta + \tau_e + \tau\right)}{\underline{\delta}\left(\bar{\delta}\tau_\eta + \tau_e + \tau\right)} = \frac{1}{\underline{\chi}}$$

which implies that

$$\frac{\partial\bar{\chi}}{\partial\tau} = \frac{\bar{\delta}\left(\bar{\delta}-\underline{\delta}\right)\tau_\eta}{\underline{\delta}\left(\bar{\delta}\tau_\eta+\tau_e+\tau\right)^2} > 0, \quad \frac{\partial\bar{\chi}}{\partial\tau_\eta} = -\frac{\bar{\delta}\left(\bar{\delta}-\underline{\delta}\right)(\tau_e+\tau)}{\underline{\delta}\left(\bar{\delta}\tau_\eta+\tau_e+\tau\right)^2} < 0, \quad \frac{\partial\bar{\chi}}{\partial\tau_e} = \frac{\bar{\delta}\left(\bar{\delta}-\underline{\delta}\right)\tau_\eta}{\underline{\delta}\left(\bar{\delta}\tau_\eta+\tau_e+\tau\right)^2} > 0. \tag{34}$$

A.6 Lemma 3 and its proof

Lemma 3. *The experienced utility is the expected utility that the player incurs under the objective distribution and is given by*

$$EU_i\left(\delta_{\eta,i}\right) = L_2 + u_{kk}\left(\frac{\kappa_1\left(1-r\right)}{1-rA}\right)^2\left(\frac{\tau+\tau_e+\delta_{\eta,i}^2\tau_\eta}{\left(\tau+\tau_e+\delta_{\eta,i}\tau_\eta\right)^2}\right). \tag{35}$$

where L_2 does not depend upon player i's beliefs.

Proof. Applying equation (30) to objective beliefs, we get $EU_i\left(\delta_{\eta,i}\right) = \mathbb{E}\left[u\left(k_i\left(\delta_{\eta,i}\right), K, \theta, \sigma\right)\right]$. Note that the only term which depends upon player i's beliefs is

$$\mathbb{E}\left[\left(k-\mathcal{K}\right)^2\right] = \left(\frac{\kappa_1\left(1-r\right)}{1-rA}\right)^2\left(\frac{\tau+\tau_e+\delta_{\eta,i}^2\tau_\eta}{\left(\tau+\tau_e+\delta_{\eta,i}\tau_\eta\right)^2}\right)$$

since it is dependent upon her action $k_i\left(\delta_{\eta,i}\right)$. This immediately implies (35). □

A.7 Proof of Proposition 2 and Corollary 3

Each player maximizes (6), which by (33) and (35) is equivalent to maximizing

$$OU\left(B_i, \chi, B\right) \equiv -\frac{1}{\tau+\tau_e+\delta_{\eta,i}\tau_\eta} + \frac{\chi B^2}{\delta_{\eta,i}\tau_\eta} - \psi\frac{\tau+\tau_e+\delta_{\eta,i}^2\tau_\eta}{\left(\tau+\tau_e+\delta_{\eta,i}\tau_\eta\right)^2} \tag{36}$$

$$= -\frac{1-B_i}{\tau+\tau_e}\left(1-\frac{\chi B^2}{B_i}\right) - \frac{\psi}{\tau+\tau_e}\left[(1-B_i)^2 + B_i^2\left(\frac{1-B_{RE}}{B_{RE}}\right)\right] \tag{37}$$

where $B_i \equiv \frac{\delta_{\eta,i}\tau_\eta}{\tau+\tau_e+\delta_{\eta,i}\tau_\eta}$ denotes the weight player i chooses to place on s. This follows since player i takes $-u_{kk}\left(\frac{\kappa_1(1-r)}{1-rA}\right)^2 > 0$ as given. Then,

$$\frac{\partial OU_i}{\partial B_i} = \frac{1}{B_i^2(\tau+\tau_e)}\left[\underbrace{\left(1+2\psi\left[1-\frac{B_i}{B_{RE}}\right]\right)B_i^2 - \chi B^2}_{\equiv f(B_i,B,B_{RE},\chi,\psi)}\right]. \tag{38}$$

Whether OU_i is increasing or decreasing in player i's subjective belief, B_i depends upon the sign of $f(B_i, B, B_{RE}, \chi, \psi)$ defined in equation (38). Note that $\frac{\partial f}{\partial B_i} = -\frac{6\psi}{B_{RE}}B_i^2 + 2(2\psi+1)B_i$ and equals zero at $B_i = 0$ and $B_i = \frac{B_{RE}(2\psi+1)}{3\psi}$. Moreover, $\frac{\partial^2 f}{\partial B_i^2}$ is positive at $B_i = 0$ (i.e., it is a minimum) and negative at $B_i = \frac{B_{RE}(2\psi+1)}{3\psi}$ (i.e., it is a maximum). This implies that, if $\frac{B_{RE}(2\psi+1)}{3\psi} < 1$, $f(B_i, B, B_{RE}, \chi, \psi)$ is increasing in the range $B_i \in (0, \frac{B_{RE}(2\psi+1)}{3\psi})$ and decreasing for higher B_i. If $\frac{B_{RE}(2\psi+1)}{3\psi} > 1$, $f(B_i, B, B_{RE}, \chi, \psi)$ is increasing in the range $B_i \in (0, 1)$.

When $\chi \leq 0$, the objective function is increasing at $B_i = 0$ since $f(0, B, B_{RE}, \chi, \psi) = -\chi B^2 > 0$.Given our observations above, this gives rise to two possibilities. Either the objective function is (i) always increasing for any feasible B_i or (ii) there is an interior, global maximum (when $f(B_i, B, B_{RE}, \chi, \psi) = 0$ for some $B_i < 1$.)

When $\chi > 0$, the objective function is decreasing at $B_i = 0$. This implies that there are three possible cases. Either the objective function is (i) always decreasing, (ii) U-shaped, or (iii) has an interior (local) maximum. The latter two cases arise when $\exists$ some B_0 such that $f(B_0, B, B_{RE}, \chi, \psi) > 0$ and $0 < B_0 \leq 1$. Case (ii) occurs if $f(B_i, B, B_{RE}, \chi, \psi) > 0$ for all feasible $B_i \geq B_0$; otherwise, case (iii) occurs.

Given this, we can establish conditions for the different equilibria over the interval $B_i \in \left[\underline{B}, \bar{B}\right]$. First, note that there is, at most, a single interior maximum. Second, both $\underline{B}$ and $\bar{B}$ are potential global maxima. Third, there are only two feasible mixed strategy equilibria: a player can mix between $\underline{B}$ and either (i) $\bar{B}$ or (ii) an interior maximum, derived below.

Step 1: Characterize the mixed-strategy equilibria

There are two cases to consider. Suppose that $f\left(\bar{B}, B, B_{RE}, \chi, \psi\right) \geq 0$. In this setting, the objective function is (weakly) increasing at the upper bound. As a result, the objective function is either upward-sloping or U-shaped, i.e., there are no interior maxima by the argument laid out above that the f function is hump-shaped. To determine the conditions on χ for each type of equilibrium, we will directly compare the expected utility of the two possible choices, $\underline{B}$ and $\bar{B}$, given the behavior of all other players, B. In a mixed-strategy equilibrium, each player must be indifferent between choosing $\underline{B}$ and $\bar{B}$, i.e.,

$$OU(\underline{B}, \chi, B) = OU(\bar{B}, \chi, B) \tag{39}$$

Simplifying this utilizing (37) yields

$$\chi B^2 = \left(\psi\left[2-\left(\frac{\bar{B}+\underline{B}}{B_{RE}}\right)\right]+1\right)\underline{B}\bar{B}. \tag{40}$$

This implies that in any mixed-strategy equilibrium utilizing $\underline{B}$ and $\bar{B}$

$$B = \sqrt{\frac{\left(\psi\left[2-\left(\frac{\bar{B}+\underline{B}}{B_{RE}}\right)\right]+1\right)\underline{B}\bar{B}}{\chi}} \text{ and } \lambda = \frac{B-\underline{B}}{\bar{B}-\underline{B}}, \tag{41}$$

where λ denotes the measure of players who choose $\bar{B}$. Note that B is decreasing in χ. Since it must be the case that $\underline{B} \leq B \leq \bar{B}$, we can substitute these bounds on B into the indifference condition to yield the bounds on χ such that this mixed equilibrium can be sustained:

$$\left(\psi\left[2-\left(\frac{\bar{B}+\underline{B}}{B_{RE}}\right)\right]+1\right)\frac{\bar{B}}{\underline{B}} > \chi > \left(\psi\left[2-\left(\frac{\bar{B}+\underline{B}}{B_{RE}}\right)\right]+1\right)\frac{\underline{B}}{\bar{B}}. \tag{42}$$

It remains, however, for us to confirm our conjecture that $f\left(\bar{B}, B, B_{RE}, \chi, \psi\right) \geq 0$. Given the equilibrium B specified above, this is true as long as

$$\left(1+2\psi\left[1-\frac{\bar{B}}{B_{RE}}\right]\right)\bar{B}^2 \geq \chi B^2 \iff \psi \leq \frac{B_{RE}}{2\bar{B}+\underline{B}-2B_{RE}}. \tag{43}$$

On the other hand, suppose that $f\left(\bar{B}, B, B_{RE}, \chi, \psi\right) < 0$. Under this conjecture, the objective function is decreasing at the upper bound and so the only feasible mixed-strategy equilibrium is where players are indifferent between $\underline{B}$ and an interior maximum, which we will denote as B_η. Utilizing the same steps as above, indifference between $\underline{B}$ and B_η requires

$$\chi B^2 = \left(\psi\left[2-\left(\frac{B_\eta+\underline{B}}{B_{RE}}\right)\right]+1\right)\underline{B}B_\eta. \tag{44}$$

This yields expressions analogous to those found in (41) for the equilibrium B and λ and analogous cutoffs for χ as those found in (42) with B_η replacing $\bar{B}$. It remains, however, to identify the equilibrium B_η which arises in this equilibrium. Note that, for an interior maximum, the first-order condition must hold given the equilibrium B. This implies that $f\left(B_\eta, B, B_{RE}, \chi, \psi\right) = 0$ is true if and only if

$$\left(1+2\psi\left[1-\frac{B_\eta}{B_{RE}}\right]\right)B_\eta^2 = \left(\psi\left[2-\frac{B_\eta+\underline{B}}{B_{RE}}\right]+1\right)\underline{B}B_\eta \iff B_\eta = \frac{B_{RE}\left(1+2\psi\right)-\psi\underline{B}}{2\psi}. \tag{45}$$

It is easy to verify that the SOC holds at $B_i = B_\eta$. Confirming our conjecture that $f\left(\bar{B}, B, B_{RE}, \chi, \psi\right) < 0$ is equivalent to showing that $B_\eta < \bar{B}$ given our observations on the properties of $f\left(\cdot\right)$. Using (45), this is the case only if

$$\psi > \frac{B_{RE}}{2\bar{B}+\underline{B}-2B_{RE}}, \tag{46}$$

which exactly corresponds to our threshold from Case 1 (given in equation (43)). Moreover, when $\psi = \frac{B_{RE}}{2\bar{B}+\underline{B}-2B_{RE}}$, $B_\eta = \bar{B}$ so that both the upper and lower bounds of the two types of mixed-strategy equilibria are continuous.

Finally, feasibility also requires that B_η be greater than or equal to $\underline{B}$. If $3\underline{B} \leq 2B_{RE}$, then this is true for any $\psi > 0$. Otherwise, it must be the case that

$$\psi \leq \frac{B_{RE}}{3\underline{B}-2B_{RE}}. \tag{47}$$

If $3\underline{B} > 2B_{RE}$, for ψ above this cutoff, mixed equilibrium doesn't exist.

Step 2: Characterize the possible symmetric equilibria

There are only three possible symmetric equilibria: everyone chooses $\underline{B}$, $\bar{B}$, or an interior maximum which we denote B_s. For B_s to be an equilibrium, the first-order condition must hold when everyone else chooses B_s, i.e., $f(B_s, B_s, B_{RE}, \chi, \psi) = 0$. This is true as long as

$$\left(1 + 2\psi\left[1 - \frac{B_s}{B_{RE}}\right]\right) B_s^2 = \chi B_s^2 \implies B_s = B_{RE}\left[\frac{1 + 2\psi - \chi}{2\psi}\right]. \tag{48}$$

Note that feasibility requires that

$$B_s < \bar{B} \iff 2\psi(\bar{B} - B_{RE}) > B_{RE}(1 - \chi) \tag{49}$$

$$B_s > \underline{B} \iff 2\psi(\underline{B} - B_{RE}) < B_{RE}(1 - \chi) \tag{50}$$

Placing weight B_s on the public signal is equivalent to choosing

$$\delta_{\eta,i} = \frac{2\psi + 1 - \chi}{2\psi - \frac{\tau_\eta}{\tau + \tau_e}(1 - \chi)} \equiv \delta_s \tag{51}$$

To ensure that this is a local maximum (we will derive conditions under which it is the global maximum below), the second-order condition must also hold, i.e., $\frac{\partial^2}{\partial B_i^2} OU(B_s, \chi, B_s) < 0$. This condition can be written as

$$-\frac{\tau_\eta(-3\chi + 2\psi + 1)(2\psi(\tau_e + \tau) + (\chi - 1)\tau_\eta)^4}{8\psi^3(\tau_e + \tau)^3(-\chi + 2\psi + 1)(\tau_e + \tau_\eta + \tau)^3} < 0 \tag{52}$$

which is true if either (i) $\chi > 2\psi + 1$ or (ii) $\chi < \frac{2\psi+1}{3}$. Note that if the former cutoff holds, then $B_s < 0$ and so the latter will be the relevant constraint.

Step 3: Establish uniqueness for the mixed strategy equilibria

To establish uniqueness, we will show that under the conditions which give rise to the mixed-strategy equilibrium (derived in step 1), no symmetric equilibrium can exist. We start with the case where $\psi \leq \frac{B_{RE}}{2\bar{B} + \underline{B} - 2B_{RE}}$ where the mixed-strategy equilibrium is mixing between $\underline{B}$ and $\bar{B}$.

Conjecture that everyone chooses $\underline{B}$. To show that this cannot be an equilibrium in this region, note that player i finds it profitable to deviate to $\bar{B}$ when $OU\left(\bar{B}, \chi, \underline{B}\right) > OU\left(\underline{B}, \chi, \underline{B}\right)$, which is true if $\chi < \left(\psi\left[2 - \left(\frac{\bar{B}+\underline{B}}{B_{RE}}\right)\right] + 1\right)\frac{\bar{B}}{\underline{B}}$. This corresponds to the upper bound for the mixed strategy equilibrium found in (42), i.e., everyone choosing $\underline{B}$ cannot be an equilibrium in this region. Similarly, everyone choosing $\bar{B}$ cannot be an equilibrium in this region since $OU\left(\underline{B}, \chi, \bar{B}\right) > OU\left(\bar{B}, \chi, \bar{B}\right)$ if $\chi > \left(\psi\left[2 - \left(\frac{\bar{B}+\underline{B}}{B_{RE}}\right)\right] + 1\right)\frac{\underline{B}}{\bar{B}}$, the lower bound in (42).

Lastly, note that player i finds it beneficial to deviate from B_s to $\underline{B}$ if $OU\left(\underline{B}, \chi, B_s\right) > OU\left(B_s, \chi, B_s\right)$, which is the case when $\chi > \psi\frac{\underline{B}}{B_{RE}}$. The symmetric equilibrium cannot arise then as long as this threshold is below the lower bound found in (42), i.e.,

$$\psi\frac{\underline{B}}{B_{RE}} < \left(\psi\left[2 - \left(\frac{\bar{B} + \underline{B}}{B_{RE}}\right)\right] + 1\right)\frac{\underline{B}}{\bar{B}} \iff \psi < \frac{B_{RE}}{2\bar{B} + \underline{B} - 2B_{RE}} \tag{53}$$

which defines the cutoff for this mixed-strategy equilibrium. Together, this implies that the conjectured mixed-strategy equilibrium when $\psi \leq \frac{B_{RE}}{2\bar{B} + \underline{B} - 2B_{RE}}$ is unique.

We next consider the case where $\psi > \frac{B_{RE}}{2\bar{B} + \underline{B} - 2B_{RE}}$, i.e., where the mixed-strategy equilibrium features mixing between $\underline{B}$ and B_η. The steps are isomorphic to those above.

First, player i deviates and places weight B_η on s when others choose $\underline{B}$ as long as $\chi < \left(\psi\left[2-\left(\frac{B_\eta+\underline{B}}{B_{RE}}\right)\right]+1\right)\frac{B_\eta}{\underline{B}}$, which is the upper bound for this equilibrium. Second, player i deviates and places weight $\underline{B}$ on s when others choose $\bar{B}$ if $\chi > \left(\psi\left[2-\left(\frac{\bar{B}+\underline{B}}{B_{RE}}\right)\right]+1\right)\frac{\underline{B}}{\bar{B}}$, which is less than the lower bound for this equilibrium.

Lastly, note that we can rewrite the lower bound for the mixed-strategy equilibrium when $\psi > \frac{B_{RE}}{2\bar{B}+\underline{B}-2B_{RE}}$ by substituting in the equilibrium B_η. This yields the following lower bound: $\chi \geq \psi\frac{\underline{B}}{B_{RE}}$ for the conjectured mixed-strategy equilibrium to arise. But this rules out the symmetric equilibrium B_s because, as we noted above, player i would deviate to $\underline{B}$ if everyone else chooses B_s in this region. Together, this implies that the conjectured mixed-strategy equilibrium when $\psi > \frac{B_{RE}}{2\bar{B}+\underline{B}-2B_{RE}}$ is unique.

Step 4: Establish existence and uniqueness for the symmetric equilibria

We start by defining the relative utility player i receives from two choices of B_i (B_1 and B_2), given χ and B (the aggregate weight placed by others):

$$\Delta\left(B_1, B_2, \chi, B\right) \equiv \int_{B_1}^{B_2} \frac{\partial OU}{\partial B_i} dB_i. \tag{54}$$

If $\Delta\left(B_{1,2}, \chi, B\right) = 0$, then player i's payoff from B_1 and B_2 is the same, i.e., she is indifferent between choosing B_1 and B_2 given χ and B. If $\Delta\left(B_1, B_2, \chi, B\right) > 0$, she prefers B_2; if $\Delta\left(B_1, B_2, \chi, B\right) < 0$, she prefers B_1. We will construct the existence and uniqueness of the symmetric equilibrium in a piecemeal fashion.

1. Suppose that $\psi \leq \frac{B_{RE}}{2\bar{B}+\underline{B}-2B_{RE}}$ and $\chi > \left(\psi\left[2-\left(\frac{\bar{B}+\underline{B}}{B_{RE}}\right)\right]+1\right)\frac{\bar{B}}{\underline{B}}$. First, we show that a symmetric equilibrium in which all players choose $\underline{B}$ exists under these conditions. Note that when $\chi = \left(\psi\left[2-\left(\frac{\bar{B}+\underline{B}}{B_{RE}}\right)\right]+1\right)\frac{\bar{B}}{\underline{B}}$, the analysis in Step 1 above implies that players are indifferent between $\underline{B}$ and $\bar{B}$ and that $\lambda = 0$ which implies $B = \underline{B}$ and so $\Delta\left(\underline{B}, \bar{B}, \chi, \underline{B}\right) = 0$. Moreover, this implies that $\Delta\left(\underline{B}, \hat{B}, \chi, \underline{B}\right) < 0$ if $\underline{B} < \hat{B} < \bar{B}$. Second, note that (38) implies that an increase in χ uniformly decreases $\frac{\partial OU}{\partial B_i}$. Thus, for any $\chi > \left(\psi\left[2-\left(\frac{\bar{B}+\underline{B}}{B_{RE}}\right)\right]+1\right)\frac{\bar{B}}{\underline{B}}$, it must be the case that $\Delta\left(\underline{B}, \hat{B}, \chi, \underline{B}\right) < 0$ for any feasible $\hat{B}$ not equal to $\underline{B}$. That is, a symmetric equilibrium in which everyone chooses $\underline{B}$ exists in this region. Moreover, the arguments found above in Step 3 rule out potential symmetric equilibria in which all players choose $\bar{B}$ or B_s in this region, establishing uniqueness.

2. Suppose that $\psi \leq \frac{B_{RE}}{2\bar{B}+\underline{B}-2B_{RE}}$ and $\chi < \left(\psi\left[2-\left(\frac{\bar{B}+\underline{B}}{B_{RE}}\right)\right]+1\right)\frac{\underline{B}}{\bar{B}}$. We show that a symmetric equilibrium in which all players choose $\bar{B}$ exists under these conditions. If $\chi = \left(\psi\left[2-\left(\frac{\bar{B}+\underline{B}}{B_{RE}}\right)\right]+1\right)\frac{\underline{B}}{\bar{B}}$, the analysis in Step 1 above implies that players are indifferent between $\underline{B}$ and $\bar{B}$ and that $\lambda = 1$ which implies $B = \bar{B}$ and so $\Delta\left(\underline{B}, \bar{B}, \chi, \bar{B}\right) = 0$. Moreover, this implies that $\Delta\left(\hat{B}, \bar{B}, \chi, \bar{B}\right) > 0$ if $\underline{B} < \hat{B} < \bar{B}$. Again, (38) implies that a decrease in χ uniformly increases $\frac{\partial OU}{\partial B_i}$. Thus, for any $\chi < \left(\psi\left[2-\left(\frac{\bar{B}+\underline{B}}{B_{RE}}\right)\right]+1\right)\frac{\underline{B}}{\bar{B}}$, it must be the case that $\Delta\left(\hat{B}, \bar{B}, \chi, \bar{B}\right) > 0$ for any feasible $\hat{B}$ not equal to $\bar{B}$. That is, a symmetric

equilibrium in which everyone chooses $\bar{B}$ exists in this region. The arguments found above in Step 3 rule out potential symmetric equilibria in which all players choose $\underline{B}$. Moreover, under these conditions on ψ and χ, $B_s > \bar{B}$ by (49) and so it is not a feasible equilibrium. This establishes uniqueness.

3. Suppose that either (i) $3\underline{B} \leq 2B_{RE}$ and $\psi > \frac{B_{RE}}{2\bar{B}+\underline{B}-2B_{RE}}$ or (ii) $3\underline{B} > 2B_{RE}$ and $\frac{B_{RE}}{3\underline{B}-2B_{RE}} \geq \psi > \frac{B_{RE}}{2\bar{B}+\underline{B}-2B_{RE}}$. Then when $\chi = \left(\psi\left[2-\left(\frac{B_\eta+\underline{B}}{B_{RE}}\right)\right]+1\right)\frac{B_\eta}{\underline{B}}$, by step 1 above, players are indifferent between $\underline{B}$ and B_η. For any χ above this threshold, the unique symmetric equilibrium is $\underline{B}$. Establishing this follows the same logic as above. For any $\chi > \left(\psi\left[2-\left(\frac{B_\eta+\underline{B}}{B_{RE}}\right)\right]+1\right)\frac{B_\eta}{\underline{B}}$, it must be the case that $\Delta\left(\underline{B}, \hat{B}, \chi, \underline{B}\right) < 0$ for any feasible $\hat{B} \neq \underline{B}$: thus, a symmetric equilibrium in which everyone chooses $\underline{B}$ exists in this region. Moreover, the arguments found above in Step 3 rule out potential symmetric equilibria in which all players choose $\bar{B}$ or B_s in this region, establishing uniqueness.

4. Suppose that either (i) $3\underline{B} \leq 2B_{RE}$ and $\psi > \frac{B_{RE}}{2\bar{B}+\underline{B}-2B_{RE}}$ or (ii) $3\underline{B} > 2B_{RE}$ and $\frac{B_{RE}}{3\underline{B}-2B_{RE}} \geq \psi > \frac{B_{RE}}{2\bar{B}+\underline{B}-2B_{RE}}$. For $\chi < \left(\psi\left[2-\left(\frac{B_\eta+\underline{B}}{B_{RE}}\right)\right]+1\right)\frac{\underline{B}}{B_\eta}$ (which simplifies to $\chi < \psi\frac{\underline{B}}{B_{RE}}$ by substituting B_η from equation 45), symmetric equilibrium in which everyone chooses $\underline{B}$ cannot exist under these conditions by the arguments in Step 3. Thus, for any χ below this threshold, the unique symmetric equilibrium is either B_s or $\bar{B}$.

First, we establish the existence of a symmetric equilibrium in which everyone chooses B_s. Under the conjectured conditions on ψ and χ above, $\chi < \frac{2\psi+1}{3}$, so that the second-order condition holds, i.e, B_s is a local maximum. If everyone chooses B_s, the only possible profitable deviation is to $\underline{B}$ but that is ruled out since $\chi < \psi\frac{\underline{B}}{B_{RE}}$ as discussed in Step 3. Thus, the optimal choice for player i is to choose B_s when others choose B_s, i.e., it is a global maximum. It is feasible only when $\underline{B} \leq B_s \leq \bar{B}$. Equation (50) always holds under these conditions and so $B_s > \underline{B}$; however, (49) implies that $B_s \leq \bar{B}$ if and only if

$$\chi \geq 2\psi + 1 - \frac{2\psi\bar{B}}{B_{RE}}. \tag{55}$$

That $B_s = \bar{B}$ is an equilibrium at this cutoff implies that $\Delta\left(\hat{B}, \bar{B}, \chi = 2\psi+1-\frac{2\psi\bar{B}}{B_{RE}}, \bar{B}\right) > 0$ for all $\hat{B} < \bar{B}$. A decrease in χ uniformly increases $\frac{\partial OU}{\partial B_i}$ and so for all χ failing to satisfy (55) and $\hat{B} < \bar{B}$, $\Delta\left(\hat{B}, \bar{B}, \chi, B_{RE}, \bar{B}\right) > 0$: $\bar{B}$ is an equilibrium.

In this region, however, it cannot be that $\bar{B}$ is an equilibrium above the cutoff in (55). For $\bar{B}$ to be an equilibrium, it must be that $f\left(\bar{B}, \bar{B}, B_{RE}, \chi, \psi\right) \geq 0$. This is only the case as long as

$$\chi \leq 2\psi + 1 - \frac{2\psi\bar{B}}{B_{RE}}. \tag{56}$$

Together, this establishes uniqueness since the two thresholds are the same: if (56) holds, the only equilibrium is when everyone chooses $\bar{B}$; if $2\psi+1-\frac{2\psi\bar{B}}{B_{RE}} < \chi < \psi\frac{\underline{B}}{B_{RE}}$, the only equilibrium is when everyone chooses B_s.

5. Suppose that $3\underline{B} > 2B_{RE}$ and $\psi > \frac{B_{RE}}{3\underline{B}-2B_{RE}}$. Under these conditions, no mixed-strategy

equilibrium exists. First, we establish the existence of a symmetric equilibrium in which everyone chooses B_s. When everyone chooses B_s, no one deviates to $\underline{B}$ as long as

$$\chi < 2\psi + 1 - \frac{2\psi \underline{B}}{B_{RE}}. \tag{57}$$

If this inequality holds and given the conjecture that $\psi > \frac{B_{RE}}{3\underline{B}-2B_{RE}}$, then $\chi < \frac{2\psi+1}{3}$, i.e., the second-order condition holds. Thus, a symmetric equilibrium in which everyone chooses B_s exists.

Next, we establish the existence of a symmetric equilibrium in which everyone chooses $\underline{B}$. Equation (48) implies that $B_s = \underline{B}$ if and only if $\chi = 2\psi + 1 - \frac{2\psi \underline{B}}{B_{RE}}$. Following the same logic as above, it is straightforward to show that $\underline{B}$ must be an equilibrium for χ above this cutoff. To show that it is only an equilibrium above this cutoff, note that $f\left(\underline{B}, \underline{B}, B_{RE}, \chi, \psi\right) < 0$ only if $\chi > 2\psi + 1 - \frac{2\psi \bar{B}}{B_{RE}}$.

Finally, establishing the cutoff between the unique symmetric equilibrium in which everyone chooses B_s and the unique equilibrium in which everyone chooses $\bar{B}$ follows the same steps as above.

A.8 Proof of Proposition 3

In the symmetric equilibrium in which all players choose $\delta_{\eta,i} = \delta^*$, the CG coefficient at the individual level is

$$CG_i = \frac{\operatorname{cov}\left(\theta - k_i, k_i\right)}{\operatorname{var}\left(k_i\right)} = \frac{\left(1-\alpha-\beta\right)\left(\alpha+\beta\right)\frac{1}{\tau} - \frac{\alpha^2}{\tau_e} - \frac{\beta^2}{\tau_\eta}}{\left(\alpha+\beta\right)^2 \frac{1}{\tau} + \frac{\alpha^2}{\tau_e} + \frac{\beta^2}{\tau_\eta}}. \tag{58}$$

Moreover, $CG_i < 0 \iff \tau_e \rho + \tau_\eta \left(1+\rho\right)^2 \delta^* \left(1-\delta^*\right) < 0$. The CG coefficient at the aggregate level is

$$CG_a = \frac{\operatorname{cov}\left(\theta - K, K\right)}{\operatorname{var}\left(K\right)} = \frac{\left(1-\alpha-\beta\right)\left(\alpha+\beta\right)\frac{1}{\tau} - \beta^2 \frac{1}{\tau_\eta}}{\left(\alpha+\beta\right)^2 \frac{1}{\tau} + \beta^2 \frac{1}{\tau_\eta}} \tag{59}$$

Moreover, $CG_a > 0 \iff \tau_e + \left(1+\rho\right)\tau_\eta \delta^* \left(1-\delta^*\right) > 0$.

A.9 Proof of Proposition 4

In a symmetric equilibrium, all players have the same interpretation of the public signal (i.e., $\delta_{\eta,i} = \delta^*$ is the same). This implies that the dispersion of CG coefficients is zero in symmetric equilibrium.

In a mixed equilibrium, a fraction λ of players choose $\delta_{\eta,i} = \delta_{opt}$, while the remaining fraction of players choose $\delta_{\eta,i} = \underline{\delta}$. This implies that the individual CG coefficients can take one of two values:

$$CG_{i,1} = \frac{\left(1-\alpha_1-\beta_1\right)\left(\alpha_1+\beta_1\right)\frac{1}{\tau} - \frac{\alpha_1^2}{\tau_e} - \frac{\beta_1^2}{\tau_\eta}}{\left(\alpha_1+\beta_1\right)^2 \frac{1}{\tau} + \frac{\alpha_1^2}{\tau_e} + \frac{\beta_1^2}{\tau_\eta}} \qquad CG_{i,2} = \frac{\left(1-\alpha_2-\beta_2\right)\left(\alpha_2+\beta_2\right)\frac{1}{\tau} - \frac{\alpha_2^2}{\tau_e} - \frac{\beta_2^2}{\tau_\eta}}{\left(\alpha_2+\beta_2\right)^2 \frac{1}{\tau} + \frac{\alpha_2^2}{\tau_e} + \frac{\beta_2^2}{\tau_\eta}}$$

This implies that the dispersion of CG coefficients is $\lambda(1-\lambda)(CG_{i,1} - CG_{i,2})^2 > 0$. Note that the individual CG coefficients don't depend on γ. As γ increases, λ increases (from equation (18)), and hence the effect on the dispersion of CG coefficients is hump-shaped.

A.10 Proof of Proposition 5

In a symmetric equilibrium, the regression coefficient of forecast error on public information is given by:

$$b_{s,i} = \frac{Cov\left(\theta - k_i, s\right)}{var\left(s\right)} = \frac{\left(1-\alpha-\beta\right)\frac{1}{\tau} - \beta\frac{1}{\tau_\eta}}{\frac{1}{\tau}+\frac{1}{\tau_\eta}} \propto (1+\rho)(1-\delta^*) \tag{60}$$

Hence, $b_{s,i} < 0 \iff \delta^* > 1$. Next,

$$b_{f,i} = \frac{Cov\left(\theta - k_i, K\right)}{var\left(K\right)} = \frac{\left(1-\alpha-\beta\right)\left(\alpha+\beta\right)\frac{1}{\tau} - \beta^2\frac{1}{\tau_\eta}}{\frac{(\alpha+\beta)^2}{\tau}+\frac{\beta^2}{\tau_\eta}} \tag{61}$$

$$\propto \tau_e + \left(1+\rho\right)\tau_\eta\delta_\eta\left(1-\delta_\eta\right) \tag{62}$$

Hence, $b_{f,i} > 0 \iff \tau_e + \left(1+\rho\right)\tau_\eta\delta^*\left(1-\delta^*\right) > 0$.

A.11 Proof of Proposition 6

The dispersion of beliefs is given by

$$\sigma_k^2 = \mathbb{E}\left[\mathrm{var}[k_i|\{\theta,\eta\}]\right] = \mathbb{E}\left[\mathrm{var}[\alpha_i s_i + \beta_i s|\{\theta,\eta\}]\right].$$

In a symmetric equilibrium, dispersion is $\sigma_k^2 = \frac{\alpha^2}{\tau_\epsilon} = \left(\frac{A}{1+\rho-\rho A}\right)^2\frac{1}{\tau_\epsilon}$.

- If players have rational expectations, A decreases as public information precision increases and hence $\frac{\partial\sigma_{k,RE}^2}{\partial\tau_\eta} < 0$.
- Dispersion in forecasts is higher than under rational expectations (i.e., $\sigma_k^2 > \sigma_{k,RE}^2$) if and only if $A > A_{RE} \iff \frac{\tau}{\tau+\tau_e+\delta^*\tau_\eta} > \frac{\tau_e}{\tau+\tau_e+\tau_\eta} \iff \delta^* < 1 \iff \gamma < -1$.
- In a symmetric equilibrium in which every one chooses $\delta_\eta = \bar{\delta}$ or $\underline{\delta}$, then $\frac{\partial\sigma_k^2}{\partial\tau_\eta} < 0$. In a symmetric equilibrium in which every one chooses $\delta_\eta = \delta_{sym}$, then

$$\frac{\partial\sigma_k^2}{\partial\tau_\eta} \propto \frac{(1+\rho)}{(1+\rho-\rho A)^2}\frac{\partial}{\partial\tau_\eta}\left(\frac{\tau_e}{\tau+\tau_e+\delta_{sym}\tau_\eta}\right) \propto \frac{\partial}{\partial\tau_\eta}\left(\frac{\tau_e}{\tau+\tau_e+\frac{2\psi+1-\chi}{2\psi-\frac{(1-\chi)\tau_\eta}{\tau_e+\tau}}\tau_\eta}\right) < 0$$

In a mixed equilibrium,

$$\sigma_k^2 = \frac{\lambda\alpha_{opt}^2 + (1-\lambda)\,\underline{\alpha}^2}{\tau_e} + \lambda\left(1-\lambda\right)\left(\frac{\left(\alpha_{opt}-\underline{\alpha}+\beta_{opt}-\underline{\beta}\right)^2}{\tau} + \frac{\left(\beta_{opt}-\underline{\beta}\right)^2}{\tau_\eta}\right) \tag{63}$$

where $\alpha_{opt} = \frac{A_{opt}}{1+\rho-\rho A}$, $\underline{\alpha} = \frac{\underline{A}}{1+\rho-\rho A}$, $\beta_{opt} = \frac{\rho B+B_{opt}}{1+\rho-\rho A}$, $\underline{\beta} = \frac{\rho B+\underline{B}}{1+\rho-\rho A}$, $\underline{A} = \frac{\tau_e}{\tau+\tau_e+\underline{\delta}\tau_\eta}$, $\bar{A} = \frac{\tau_e}{\tau+\tau_e+\bar{\delta}\tau_\eta}$, and $A_{opt} = \frac{\tau_e}{\tau+\tau_e+\delta_{opt}\tau_\eta}$. The first term in equation (63) captures the dispersion of beliefs within the group and the second term captures the dispersion across groups. Finally,

$$\frac{\partial\sigma_k^2}{\partial\gamma} = \underbrace{\frac{\partial\lambda}{\partial\gamma}}_{>0}\left[\underbrace{\frac{\alpha_{opt}^2-\underline{\alpha}^2}{\tau_e}}_{<0} + (1-2\lambda)\underbrace{\left(\frac{\left(\alpha_{opt}-\underline{\alpha}+\beta_{opt}-\underline{\beta}\right)^2}{\tau} + \frac{\left(\beta_{opt}-\underline{\beta}\right)^2}{\tau_\eta}\right)}_{>0}\right]$$

which in turn implies that the effect of γ on σ_y^2 can be non-monotonic. Finally, the likelihood of mixed equilibrium increases in $\bar{\gamma} - \underline{\gamma}$. For $\psi = 0$, this expression simplifies to

$$\bar{\gamma} - \underline{\gamma} = \left(\frac{\bar{B}}{\underline{B}} - \frac{\underline{B}}{\bar{B}} \right)$$

which decreases in τ_η. Hence, by continuity, for ψ low enough, the likelihood of mixed equilibrium is higher when public information is less precise.

Online Appendix

B Extensions

B.1 Subjective beliefs about private information

We begin with the case when players entertain subjective beliefs about their private information but have rational beliefs about the informativeness of the public signal. From equation (11), a player's anticipatory utility is always increasing in her subjective beliefs about the private signal. This implies that there always exists a symmetric equilibrium, as characterized by the following result.

Proposition 7. *Suppose all players have rational beliefs about the public signal i.e., $\delta_{\eta,i} = 1$. There exists a symmetric equilibrium in which all players are over-confident about their private signals and choose $\delta_{e,i}$, where:*

$$\delta_{e,i} = \begin{cases} \frac{2\psi+1}{2\psi - \frac{\tau_e}{\tau+\tau_\eta}} & \text{if} \quad \psi > \frac{1+\bar{\delta}\frac{\tau_e}{\tau+\tau_\eta}}{2\left(\bar{\delta}-1\right)} \\ \bar{\delta} & \text{otherwise.} \end{cases} \tag{64}$$

Moreover, $\delta_{e,i}$ is (weakly) increasing in τ_e, and (weakly) decreasing in τ, τ_η, and ψ.

Consistent with the intuition from Section 4.2, players always want to distort their perception of private signal precision upwards. However, the experienced utility component of the objective now adds a "utility" cost to distorting beliefs. Intuitively, the more player i distorts her beliefs, the larger the expected loss in experienced utility (under the objective measure) from using subjective beliefs when choosing actions. Intuitively, this overestimation is larger when it is less costly to deviate from rational expectations (e.g., when prior information or public information is imprecise, i.e., low τ and low τ_η, or the cost of distorting beliefs is small, i.e., low ψ) and when the available private information is more precise (high τ_e). When the overall costs are sufficiently low (e.g., if ψ is sufficiently low), the player chooses the maximal distortion feasible i.e., $\delta_{e,i} = \bar{\delta}$.

Next, we allow players to entertain subjective beliefs about both their private and public signals, and assume that they maximize a weighted average of anticipatory and experienced utility. While we are not able to offer a complete characterization of equilibria for the entire parameter range, as in the earlier subsections, we establish sufficient conditions for the existence of symmetric and mixed equilibria of each type in the following results.

Proposition 8. *Suppose players choose beliefs about both private and public information and let*

$$B_{opt} = \max\left\{\min\left\{\frac{B_{RE}(2\psi+1) - \psi\underline{B}}{2\psi}, \bar{B}\right\}, \underline{B}\right\} \qquad \Gamma = 1 + \psi\left(2 - \frac{\underline{B} + B_{opt}}{B_{RE}}\right) \tag{65}$$

$$\underline{B} = \frac{\underline{\delta}\tau_\eta}{\tau + \bar{\delta}\tau_e + \underline{\delta}\tau_\eta}, \quad \bar{B} = \frac{\bar{\delta}\tau_\eta}{\tau + \bar{\delta}\tau_e + \bar{\delta}\tau_\eta}, \quad B_{RE} = \frac{\tau_\eta}{\tau + \tau_e + \tau_\eta}. \tag{66}$$

Assume $\psi < \frac{\tau_e}{2(\tau+\bar{\delta}\tau_e)}$. Then, the following hold.

(i) If $\chi > \frac{B_{opt}}{\underline{B}}\Gamma$, then there exists a unique equilibrium in which all players choose $\delta_{e,i} = \bar{\delta}$ and $\delta_{\eta,i} = \underline{\delta}$.

(ii) If $\chi < \frac{B}{B_{opt}}\Gamma$, then there exists a unique equilibrium in which all players choose $\delta_{e,i} = \bar{\delta}$ and

$$\delta_{\eta,i} = \min\left\{\bar{\delta}, \frac{2\psi + 1 - \chi}{2\psi - \frac{(1-\chi)\tau_\eta}{\bar{\delta}\tau_e + \tau}}\right\}. \tag{67}$$

(iii) For $\chi \in \left[\frac{B}{B_{opt}}\Gamma, \frac{B_{opt}}{\underline{B}}\Gamma\right]$, then there exists a unique equilibrium in which a fraction λ of players choose $\delta_{e,i} = \bar{\delta}$ and $\delta_{\eta,i} = \frac{B_{opt}}{1-B_{opt}}\frac{(\bar{\delta}\tau_e+\tau)}{\tau_\eta}$ while the remaining players choose $\delta_{e,i} = \bar{\delta}$ and $\delta_{\eta,i} = \underline{\delta}$.

The above result shares features of the equilibrium characterization in our main analysis. Specifically, the condition on the relative importance of experienced utility i.e., $\psi < \frac{\tau_e}{2(\tau+\bar{\delta}\tau_e)}$, ensures that in any equilibrium, players choose to maximally distort their beliefs about their private information i.e., set $\delta_{e,i} = \bar{\delta}$.[39] Given this, when χ is sufficiently small (part (i)), we show there exists a symmetric equilibrium in which all players exhibit over-confidence about their private information and over-react to public information. Similarly, when χ is sufficiently large (part (ii)), all players exhibit over-confidence in their private information, but under-react to public information. And for intermediate χ, there exists a unique mixed equilibrium in which some players over-react to the public signals, while others under-react to it.

B.1.1 Proof of Proposition 7

The objective of player i is

$$\max_{\delta_{e,i}\in(\underline{\delta},\bar{\delta})} AU_i + \psi EU_i$$

The lower constraint is never binding (when players only choose $\delta_{e,i}$) and the objective can be rewritten as

$$\max_{\delta_{e,i}} AU_i + \psi EU_i \quad \text{subject to } \delta_{e,i} < \bar{\delta}$$

Let ω_e denote the Lagrange multiplier for the above inequality. We define the Lagrangian

$$L = -\frac{1}{\tau + \delta_{e,i}\tau_e\tau_\eta} + \chi\frac{B^2}{\tau_\eta} - \psi\frac{\tau + \delta_{e,i}^2\tau_e + \tau_\eta}{(\tau + \delta_{e,i}\tau_e + \tau_\eta)^2} - \omega_e(\delta_{e,i} - \bar{\delta}).$$

The FOC with respect to $\delta_{e,i}$ for any equilibrium is

$$\frac{\tau_e}{(\tau_\eta + \delta_{e,i}\tau_e + \tau)^2}\left(1 - 2\psi\delta_{e,i} + \frac{2\psi\left(\tau + \delta_{e,i}^2\tau_e + \tau_\eta\right)}{\tau + \delta_{e,i}\tau_e + \tau_\eta}\right) - \omega_e = 0. \tag{68}$$

Following steps similar to those in the proof of Proposition 2, it can be shown that the objective function has either a single interior maximum or is always increasing. Hence, the

[39] While a small enough ψ is needed for the proof, numerical simulations show that the qualitative results hold more generally. Thus, we view the requirement of a small ψ in Proposition 8 as technical and not restrictive for the economic mechanism.

solution is

$$\delta_{e,i} \equiv \min\left(\frac{2\psi+1}{2\psi-\frac{\tau_e}{\tau+\tau_\eta}}, \bar{\delta}\right). \tag{69}$$

B.1.2 Proof of Proposition 8

The objective of player i is

$$\max_{\delta_{\eta,i},\delta_{e,i}\in[\underline{\delta},\bar{\delta}]} -\frac{1}{\tau+\delta_{e,i}\tau_e+\delta_{\eta,i}\tau_\eta}+\chi\frac{B^2}{\delta_{\eta,i}\tau_\eta}-\psi\frac{\tau+\delta_{e,i}^2\tau_e+\delta_{\eta,i}^2\tau_\eta}{\left(\tau+\delta_{e,i}\tau_e+\delta_{\eta,i}\tau_\eta\right)^2}.$$

Let's first focus on the choice of $\delta_{e,i}$. Let $\omega_{e,1}$ and $\omega_{e,2}$ denote the Lagrange multipliers for the inequalities $\underline{\delta} < \delta_{e,i}$ and $\delta_{e,i} < \bar{\delta}$ respectively. We define the Lagrangian

$$L = -\frac{1}{\tau+\delta_{e,i}\tau_e+\delta_{\eta,i}\tau_\eta}+\chi\frac{B^2}{\delta_{\eta,i}\tau_\eta}-\psi\frac{\tau+\delta_{e,i}^2\tau_e+\delta_{\eta,i}^2\tau_\eta}{\left(\tau+\delta_{e,i}\tau_e+\delta_{\eta,i}\tau_\eta\right)^2}+\omega_{e,1}\delta_{e,i}-\omega_{e,2}\delta_{e,i}.$$

The FOC with respect to $\delta_{e,i}$ for any equilibrium is

$$\frac{\tau_e}{\left(\delta_{\eta,i}\tau_\eta+\delta_{e,i}\tau_e+\tau\right)^2}\left(1-2\psi\delta_{e,i}+2\psi\frac{\left(\tau+\delta_{e,i}^2\tau_e+\delta_{\eta,i}^2\tau_\eta\right)}{\tau+\delta_{e,i}\tau_e+\delta_{\eta,i}\tau_\eta}\right)+\omega_{e,1}-\omega_{e,2}=0.$$

If $\psi < \frac{\tau_e}{2(\tau+\bar{\delta}\tau_e)}$, the constraint $\delta_{e,i} \leq \bar{\delta}$ is binding (irrespective of the choice of $\delta_{\eta,i}$) and all players choose $\delta_{e,i} = \bar{\delta}$. The rest of the proof follows from proposition 2 with τ_e replaced with $\bar{\delta}\tau_e$.

B.2 Subjective beliefs about others' signals

In this section, we modify our benchmark model to allow players to hold subjective beliefs about the private signals observed by other players. As in the benchmark model, each individual observes

$$s_i = \theta + \varepsilon_i \quad \varepsilon_i \sim N\left(0, 1/\tau_e\right), \tag{70}$$

but player i believes that

$$\varepsilon_j =_i \sqrt{1-p_i^2}\eta + p_i\varepsilon_j \quad \varepsilon_j \sim N\left(0, \tfrac{1}{\delta_i\tau_e}\right), \tag{71}$$

where p_i and δ_i are chosen by player i to maximize her anticipatory utility, net of costs. To highlight the role of beliefs about others, we shut down the public signal, s, and we do not allow players to hold subjective beliefs about the precision of their own signal. All other features of the benchmark model are unchanged.

Suppose player i conjectures that $K = \kappa_0 + \alpha\left(\theta + \sqrt{1-p_i^2}\eta\right)$, where α is determined in equilibrium. Then,

$$k_i = \kappa_0 + \left(r\alpha + (1-r)\kappa_1\right)\mathbb{E}_i\left[\theta|s_i\right] \tag{72}$$

This implies that, in equilibrium, player i believes that the average action (across all players) is given by

$$K = \int_i k_i di = \kappa_0 + \left(r\alpha + (1-r)\kappa_1\right)A\left(\theta + \sqrt{1-p_i^2}\eta\right), \tag{73}$$

Matching terms, we show that α is unchanged from our benchmark model. Players' subjective belief about the private signals of others is given by the following proposition.

Proposition 9. *Suppose players choose subjective beliefs about the private signals of others to maximize a weighted average of anticipatory and experienced utility (as in (6)). Then,*

(i) If $u_{\sigma\sigma} \geq u_{KK}$, all players choose $p_i = 1$. Moreover, players choose $\delta_i = \underline{\delta}$ iff $u_{\sigma\sigma} > 0$, and choose $\delta_i = \bar{\delta}$ otherwise.

(ii) If $u_{\sigma\sigma} < u_{KK}$, all players choose $p_i = 0$. Moreover, players choose $\delta_i = \underline{\delta}$ iff $u_{KK} > 0$, and choose $\delta_i = \bar{\delta}$ otherwise.

Proof. We can rewrite anticipatory utility as

$$AU_i(p_i, \delta_i) = \begin{array}{c} \mathbb{E}[u_0] + \mathcal{A}\text{var}(\theta) + \frac{1}{2}u_{\sigma\sigma}\mathbb{E}_i[\sigma_k^2] \\ +\frac{1}{2}\left(u_{kk}\mathbb{E}_i\left[(k-\mathcal{K})^2\right] + (u_{KK} - r^2 u_{kk})\,\mathbb{E}_i\left[(K-\kappa)^2\right]\right), \end{array} \tag{74}$$

where the key distinction from our benchmark is that players' subjective beliefs about other's signals impact their expectation of the dispersion in their actions, σ_k^2. Given player i's subjective beliefs,

$$\mathbb{E}_i\left[\sigma^2\right] = \int (k_i - K)^2 \, di = \frac{p_i^2\alpha^2}{\delta_i\tau_e} \tag{75}$$

$$\mathbb{E}_i\left[(k-\mathcal{K})^2\right] = \frac{(r\alpha + (1-r)\,\kappa_1)^2}{\tau} + \frac{r^2\alpha^2(1-p_i^2)}{\delta_i\tau_e} \tag{76}$$

$$\mathbb{E}_i\left[(K-\kappa)^2\right] = \frac{\left(\frac{\kappa_1(A-1)}{1-rA}\right)^2}{\tau} + \frac{\alpha^2(1-p_i^2)}{\delta_i\tau_e} \tag{77}$$

This implies that

$$\frac{\partial AU_i}{\partial \delta_i} = -u_{\sigma\sigma}\left(\frac{p_i^2\alpha^2}{2\delta_i^2\tau_e}\right) - u_{KK}\left(\frac{\alpha^2(1-p_i^2)}{2\delta_i^2\tau_e}\right) \tag{78}$$

$$= -\frac{\alpha^2}{2\delta_i^2\tau_e}\left(p_i^2 u_{\sigma\sigma} + \left(1-p_i^2\right)u_{KK}\right) \tag{79}$$

$$\frac{\partial AU_i}{\partial p_i} = u_{\sigma\sigma}\left(\frac{p_i\alpha^2}{\delta_i\tau_e}\right) - u_{KK}\frac{\alpha^2 p_i}{\delta_i\tau_e} \tag{80}$$

$$= \frac{p_i\alpha^2}{\delta_i\tau_e}\left(u_{\sigma\sigma} - u_{KK}\right) \tag{81}$$

Finally, note that since player i's action does not depend on p_i or δ_i, their experienced utility is also independent of p_i or δ_i. Hence, we have the result. □

B.3 Subjective beliefs about means

In this section, we modify our benchmark model to allow players to hold subjective beliefs about the mean of both their own private signal as well as the public signal. Specifically, player i believes the errors in her signals are

$$\varepsilon_i \sim_i N\left(\mu_{e,i}, \frac{1}{\tau_e}\right), \quad \eta \sim_i N\left(\mu_{\eta,i}, \frac{1}{\tau_\eta}\right).$$

Given player i's subjective beliefs, $\mu_{e,i}$ and $\mu_{\eta,i}$, and the realization of s_i and s, her optimal action, k_i is

$$k_i^*\left(\mu_{e,i},\mu_{\eta,i}\right)=\mathbb{E}_i\left[rK+(1-r)\,\kappa\left(\theta\right)|s_i,s\right]=r\mathbb{E}_i\left[K|s_i,s\right]+(1-r)\,\mathbb{E}_i\left[\kappa\left(\theta\right)|s_i,s\right],$$

Given our assumptions on the joint distribution of fundamentals and signals, Bayesian updating implies that player i's conditional beliefs about θ are given by

$$\mathbb{E}_i\left[\theta|s_i,s\right]=A\left(s_i-\mu_{e,i}\right)+B\left(s-\mu_{\eta,i}\right),\text{ and }\mathrm{var}_i\left[\theta|s_i,s\right]=\frac{1-A-B}{\tau},$$

where player i's weights on the private and public signals are given by:

$$A\equiv\frac{\tau_e}{\tau+\tau_e+\tau_\eta},\text{ and }B\equiv\frac{\tau_\eta}{\tau+\tau_e+\tau_\eta}.$$

With this, we establish the existence of a unique equilibrium, given players' beliefs.[40]

Lemma 4. *Given a choice of subjective beliefs $\{\mu_{e,i},\mu_{\eta,i}\}_i$ for all players, there always exists a unique, linear equilibrium in which player i's optimal action is given by:*

$$k_i\left(\mu_{e,i},\mu_{\eta,i}\right)=\kappa_0+r\gamma_e\mu_e+r\gamma_\eta\mu_\eta+r\beta\mu_{\eta,i}+\alpha\left(s_i-\mu_{e,i}\right)+\beta\left(s-\mu_{\eta,i}\right),$$

and the aggregate action is given by:

$$K=\kappa_0+\gamma_e\mu_e+\gamma_\eta\mu_\eta+\alpha\theta+\beta s,$$

where $\alpha=\frac{A(1-r)}{1-rA}\kappa_1,\beta=\frac{B}{1-rA}\kappa_1,\mu_e=\int\mu_{e,i}di,\ \mu_\eta=\int\mu_{\eta,i}di,\gamma_e=-\frac{\kappa_1 A}{1-rA},\ \gamma_\eta=-\frac{\kappa_1 B}{1-rA}$

With the equilibrium established, we then establish how $\mu_{e,i}$ and $\mu_{\eta,i}$ affect both players' anticipatory utility as well as the cost borne (through the experienced utility penalty).

Proposition 10. *Given player i's subjective beliefs, $\mu_{e,i}$ and $\mu_{\eta,i}$, anticipatory utility can be written*

$$AU_i\left(\mu_{e,i},\mu_{\eta,i}\right)=\Gamma+u_K\beta\left(\mu_{\eta,i}-\mu_\eta\right)+\frac{1}{2}\left(u_{KK}-r^2u_{kk}\right)\left(\gamma_e\mu_e+\beta\left(\mu_{\eta,i}-\mu_\eta\right)\right)^2 \tag{82}$$

and the experienced utility penalty is

$$EU_i\left(\mu_{e,i},\mu_{\eta,i}\right)=u_{kk}\left(\alpha\mu_{e,i}+\beta\left(1-r\right)\mu_{\eta,i}\right)^2$$

Players' anticipatory utility doesn't depend on $\mu_{e,i}$: intuitively, shifting the mean of the private signal is "accounted for" when updating and so player i's anticipatory utility is unaffected. However, since the true mean is zero, choosing $\mu_{e,i}\neq 0$ is objectively costly. As a result, all players choose $\mu_{e,i}=0$.

Thus, we rewrite the objective of player i:

$$\max_{\mu_{\eta,i}}AU_i+\psi EU_i.$$

First, we characterize the equilibrium when aggregate errors volatility decreases anticipatory utility, i.e., when $\chi<0$. In order to characterize the equilibrium, we assume that $\mu_{\eta,i}\in[-\bar{\mu},\bar{\mu}]$. This is similar to the set of reasonable beliefs imposed in our baseline analysis.

Proposition 11. *If aggregate errors volatility decreases anticipatory utility (i.e., $\chi\leq 0$), then there exists a unique symmetric equilibrium in which*

$$\mu_e=0\qquad\mu_{\eta,i}=\mu_\eta=\min\left(\max\left(\frac{u_K}{-2u_{kk}\psi\beta\left(1-r\right)^2},-\bar{\mu}\right),\bar{\mu}\right)\ \textit{for all } i\in[0,1]. \tag{83}$$

[40] All the proofs for this section follow similar steps as in the main model and are omitted for brevity.

Next, we consider the case when aggregate errors volatility increases anticipatory utility.

Proposition 12. *If aggregate errors volatility increases anticipatory utility (i.e., $\chi > 0$), then there exists a $\bar{\chi} > 0$ such that:*

(i) If $\chi > \bar{\chi}$, the unique equilibrium is mixed. In this equilibrium, a fraction λ of players optimally chooses $\mu_{e,i} = 0$ and $\mu_{\eta,i} = -\bar{\mu}$, while the remaining fraction $1 - \lambda$ optimally chooses $\mu_{e,i} = 0$ and $\mu_{\eta,i} = \bar{\mu}$.

(ii) If $\chi < \bar{\chi}$, then the unique equilibrium is symmetric and is given by (83).

While players never choose to distort their beliefs about the mean of their private signals, players always choose to distort their beliefs about the mean of the public signal. Specifically, we find that there exists a symmetric equilibrium when either the effect of aggregate errors volatility on anticipatory utility is not too high. In this symmetric equilibrium, whether players choose to believe that the public signal is biased upward or downward (i.e., positive or negative μ_η, respectively) depends upon whether the aggregate action, u_K, imposes a positive or a negative externality. On the other hand, when non-fundamental volatility increases anticipatory utility sufficiently, there does not exist a symmetric equilibrium - instead, players endogenously choose to disagree about the mean of the public signal.

C Applications

In this section, we consider the implications of our analysis for a few applied models.

C.1 Efficient Competitive Economies

Consider an incomplete-market competitive economy in which agents' choices are strategic substitutes. There are two goods and a continuum of households (who act as consumers and producers). Let q_{1i} and q_{2i} denote the respective quantities purchased by consumer i (the consumer living in household i). The preferences of this consumer are given by

$$u_i = \theta q_{1i} - \frac{b q_{1i}^2}{2} + q_{2i}$$

where the random variable θ represents a shock in the relative demand for the two goods and the budget is

$$p q_{1i} + q_{2i} = e + \pi_i$$

where p is the price of good one, good two serves as numeraire , e is an exogenous endowment of good 2, and π_i are the profits of producer i. Profits in turn are given by $\pi_i = p k_i - \frac{k_i^2}{2}$ where k_i denotes the quantity of good 1 produced by household i.

Consumer i chooses the optimal bundle (q_{1i}, q_{2i}) so as to maximize utility subject to budget constraint, which gives $p = \theta - b q_{1i}$. Households are ex-ante identical which, together with market clearing, implies that $q_{1i} = K$ for all i and therefore $p = \theta - bK$ where $K = \int k_i di$. This example is thus nested in our model with the utility

$$U(k, K, \sigma_k, \theta) = (\theta - bK)\, k - k^2/2 + bK^2/2 + e$$

which implies that

$$u_{kk} = -1, u_{kK} = -b, u_{k\theta} = 1, u_{KK} = b$$

and $\kappa_0 = 0; \kappa_1 = \frac{1}{1+b}$ and

$$\chi \equiv \frac{u_{KK} - u_{kk} r^2}{-u_{kk} (1-r)^2} = \frac{b}{1+b} > 0.$$

Since $\chi \in (0,1)$ increases in b, Proposition 2 shows that whether the equilibrium belief choice is symmetric or mixed depends upon the level of b. If b is too low, then the equilibrium is symmetric. For high enough b, the equilibrium is mixed. Proposition 3 implies that, in symmetric equilibrium (low b), the players overreact to public information. In a mixed equilibrium, players under-react to information if $\chi = \frac{b}{1+b} > \frac{\Gamma \underline{B} B_{opt}}{B_{RE}^2}$ i.e., if b is high. Moreover, the average weight on public information decreases in b.

Proposition 4 implies that, in a symmetric equilibrium (i.e., low b), dispersion of beliefs increases with b. In a mixed equilibrium (when b is high), the dispersion of beliefs is hump-shaped in b. Proposition 5 implies that, in a symmetric equilibrium, there is over-reaction at the individual level and there will be under-reaction at the consensus level if either the set of reasonable beliefs is small or if b is high. Moreover, both CG_i and CG_a are increasing in b. Proposition 7, implies that, when b is high, the dispersion of individual CG coefficients is hump-shaped in χ.

C.2 Investment complementarities

Consider a setting in which the terminal value of the firm is given by $V \equiv V(R,k)$ where R measures the return on investment, or productivity, of the project available and k represents the scale of investment in the project. For analytical tractability, let

$$V(R,k) = Rk - \tfrac{1}{2}k^2, \tag{84}$$

$$R = (1-a)\theta + aK, \tag{85}$$

where $\theta \in R$ represents the firm's exogenous productivity. This implies that firm i's utility is

$$U_i(k_i, K, \theta) = (1-a)\theta k_i + aKk_i - \tfrac{1}{2}k_i^2$$

which falls into the class of general objective functions we analyzed. This implies that

$$u_{kk} = -1, u_{kK} = a, u_{k\theta} = 1-a.$$

which implies that $r = a$ and $\kappa_1 = 1$ and so

$$\chi = -\frac{(u_{KK} - r^2 u_{kk})}{u_{kk}(1-r)^2} = \frac{a^2}{(1-a)^2} > 0$$

In this economy, χ increases with a. Moreover, $\chi < 1 \iff a < \frac{1}{2}$. Proposition 2 shows that whether the equilibrium belief choice is symmetric or mixed depends upon the level of a. If a is too low or too high, then the equilibrium is symmetric. For medium a, the equilibrium is mixed. Proposition 3 implies that, in symmetric equilibrium, the players under-react to public information if $a > \frac{1}{2}$. In a mixed equilibrium, players under-react to information if $\chi = \frac{a}{1-a} > \frac{\Gamma \underline{B} B_{opt}}{B_{RE}^2}$ i.e., if a is high enough. Moreover, the average weight on public information decreases in a.

Proposition 4 implies that, in a symmetric equilibrium, the dispersion of beliefs increases with a. In a mixed equilibrium, the dispersion of beliefs is hump-shaped in a. Proposition 5 implies that, in a symmetric equilibrium, there is over-reaction at the individual level iff

$a < \frac{1}{2}$ and there will be under-reaction at the consensus level if either the set of reasonable beliefs is small or if a is high. Moreover, both CG_i and CG_a are increasing in a. Proposition 7, implies that, in a mixed equilibrium, the dispersion of individual CG coefficients is hump-shaped in a.

C.3 Cournot versus Bertrand

We next study two IO applications with a large number of firms: a Cournot-like game, where firms compete in quantities and actions are strategic substitutes; and a Bertrand-like game, where firms compete in prices and actions are strategic complements.

First, consider a Cournot setting in which firms compete on quantity. For any firm, consumer demand is $p = a_0 + a_1\theta - a_2 q - a_3 Q$ (with $a_0, a_1, a_2, a_3 > 0$), where p denotes the price at which the firm sells each unit of its product, q is the quantity it produces, Q is the average quantity produced across all firms, and θ is a fundamental demand shock/shifter. Each firm's profits are given by $u = pq - C(q)$, where $C(q) = c_1 q + c_2 q^2$ is the cost function (with $c_1, c_2 > 0$). This model is nested in our general framework with $k \equiv q, K \equiv Q$, and

$$U(k, K, \sigma_k, \theta) = (a_0 - c_1 + a_1\theta - a_3 K)\, k - (a_2 + c_2)\, k^2.$$

This implies that $U_{kk} = -2(a_2 + c_2), U_{kK} = -a_3, U_{k\theta} = a_1, U_k = a_0 - c_1$ which in turn implies that $r = \frac{-a_3}{2(a_2+c_2)}$,$\kappa_1 = -\frac{a_1}{-2(a_2+c_2)-a_3}$ and

$$\chi = -\frac{(u_{KK} - r^2 u_{kk})}{u_{kk}(1-r)^2} = \frac{\frac{a_3^2}{4(a_2+c_2)^2}}{\left(1 + \frac{a_3}{2(a_2+c_2)}\right)^2} > 0$$

which implies that $\chi \in (0,1)$ and is increasing in $b \equiv \frac{a_3}{2(a_2+c_2)}$. This, in turn implies that χ increases in a_3 but decreases in a_2 and c_2.

Next, consider a Bertrand setting in which firms compete on price. Consumer demand for each firm is given by $q = b_0 + b_1\theta' - b_2 p + b_3 P$, where q denotes the quantity sold by the firm, p is the price the firm sets, P is the average price in the market, and θ' is an exogenous demand shifter ($b_0, b_1, b_2, b_3 > 0$); we naturally impose $b_3 < b_2$, so that an equal increase in p and P reduces q. Individual profits are $u = pq - C(q)$, where $C(q) = c_1 q + c_2 q^2$ (with $c_1, c_2 > 0$). This model is nested in our framework with $k \equiv p - c_1, K \equiv P - c_1$ (actions are now prices), and

$$U(k, K, \sigma_k, \theta) = \theta k(1+2c) - k^2(1+c) + Kk(b+2bc) - c\theta^2 - cb^2K^2 - 2bc\theta K$$

where $\theta = \frac{b_0}{b_2} + \frac{b_1}{b_2}\theta' - c_1(1-b)$, $b \equiv \frac{b_3}{b_2}$ and $c \equiv c_2 b_2$. This implies that $U_{kk} = -2(1+c), U_{kK} = b(1+2c), U_{k\theta} = 1+2c, U_{KK} = -2cb^2, U_{K\theta} = -2bc, U_{\theta\theta} = -2c$ which in turn implies that $r = \frac{b(1+2c)}{2(1+c)}$, $\kappa_1 = \frac{1+2c}{2(1+c)-b(1+2c)}$ and

$$\chi = -\frac{(u_{KK} - r^2 u_{kk})}{u_{kk}(1-r)^2} = \frac{b^2}{(2(1+c) - b(1+2c))^2} > 0.$$

Since $b < 1$, one can show that $\chi \in (0,1)$. Moreover, χ is increasing in $b = b_3/b_2$ and decreases in $c = c_2 b_2$. This implies that χ increases in b_3, decreases in b_2 and c_2.

C.4 Information Spillovers

The model of Angeletos, Lorenzoni, and Pavan (2018) considers a novel channel through which the information in the real sector affects behavior in the financial sector. The real

sector is comprised of entrepreneurs making investment decisions, and the financial sector is comprised of investors who provide liquidity to the "real" economy. All players are risk-neutral and the discount rate is zero. Time is divided in three periods, $t \in \{1, 2, 3\}$. At $t = 1$, a new investment opportunity becomes available with productivity $\theta \sim N\left(0, \frac{1}{\tau_\theta}\right)$. This investment pays off at $t = 3$. There is a continuum of entrepreneurs who can choose how much to invest in new technology. Let k_i denote the investment of entrepreneur i, and let the cost of this investment be $\frac{k_i^2}{2}$. Entrepreneurs have access to information technology that generates both a private and a "public" signal that they utilize when making their investment decision.[41] The joint distribution of fundamentals and signals follows the specification detailed in Section 3.

At t = 2, the "financial market" operates: some entrepreneurs transfer their capital to the traders. Each entrepreneur is hit by an idiosyncratic shock with probability $l \in [0, 1]$. Entrepreneurs hit by this shock do not value consumption at t = 3 and have no choice but to sell all their capital at t = 2, at a price p, to investors. For simplicity, the entrepreneurs not hit by the shock are precluded from trading. Entrepreneur preferences are given by $u_i = c_{i1} + c_{i2} + s_i c_{i3}$, where c_{it} denotes agent i's consumption in period t, while s_i is a random variable that takes value 0 if the agent is hit by a liquidity shock and value 1 otherwise. Each entrepreneur's expected utility at the time of investment is given by

$$\mathbb{E}_i\left[u_i | s_i, s\right] = \mathbb{E}_i\left[(1-l)\,\theta k_i + lpk_i - \frac{1}{2}k_i^2 | s_i, s\right]. \tag{86}$$

The financial market is competitive and the market-clearing price is denoted by p. Investors do not have access to their own information technology but, given the assumptions above, update their beliefs about the productivity of the technology utilizing the information contained in the supply of capital to be liquidated. It can be shown that, given the distributional assumptions and the risk-neutrality of traders, that $p = \mathbb{E}\left[\theta | K\right] = \alpha_1 K$, where α_1 is pinned down in equilibrium. This implies that expected utility of entrepreneur is given by

$$\mathbb{E}\left[u_i\left(k_i, K, \theta\right) | s_i, s\right] = \mathbb{E}_i\left[(1-l)\,\theta k_i + lk_i\alpha_1 K - \frac{1}{2}k_i^2 | s_i, s\right]. \tag{87}$$

and therefore entrepreneurs optimal action is given by $k_i = \mathbb{E}_i\left[(1-l)\,\theta + l\alpha_1 K\right]$. Note that this objective falls into the class of general objective functions we studied. Aggregating across all entrepreneurs, this implies that aggregate investment can be written as

$$K = \frac{1-l}{1-lA}\kappa_1 A\theta + \frac{B}{1-lA}\kappa_1 s \tag{88}$$

$$= \frac{\kappa_1\left((1-l)\,A + B\right)}{1-lA}\left(\theta + \frac{B}{(1-l)\,A + B}\eta\right). \tag{89}$$

It is straightforward to see that the aggregate level of capital reveals a signal of the form $\xi = \theta + \frac{B}{((1-r)A+B)}\eta$ to investors which, given the linear-normal structure, verifies the

[41] It is public in the sense that all entrepreneurs observe the same signal; however, as we discuss below, it is assumed that investors do not observe the signal and can only learn about θ through the aggregate investment level.

conjectured functional form for the price of capital. This implies that α_1 solves the equation

$$\frac{\alpha_1 (1-l)}{1-l\alpha_1} = \frac{((1-l\alpha_1) A + B) \tau_\eta}{((1-l\alpha_1) A + B)^2 \tau_\eta + B^2 \tau_\theta} (1 - l\alpha_1 A)$$

On the other hand, however, note that

$$\chi \equiv \frac{u_{KK} - u_{kk} r^2}{-u_{kk} (1-r)^2} = \frac{l^2 \alpha_1^2}{(1-l\alpha_1)^2} > 0.$$

Note that χ increases with l. Moreover, $\chi < 1 \iff l\alpha_1 < \frac{1}{2}$. Proposition 2 shows that whether the equilibrium belief choice is symmetric or mixed depends upon the level of $l\alpha_1$. If $l\alpha_1$ is too low or too high, then the equilibrium is symmetric. For medium $l\alpha_1$, the equilibrium is mixed. Proposition 3 implies that, in symmetric equilibrium, the players under-react to public information if $l\alpha_1 > \frac{1}{2}$. In a mixed equilibrium, players under-react to information if $\chi = \frac{l\alpha_1}{1-l\alpha_1} > \frac{\Gamma B B_{opt}}{B_{RE}^2}$ i.e., if l is high enough. Moreover, the average weight on public information decreases in l.

Proposition 4 implies that, in a symmetric equilibrium, the dispersion of beliefs increases with l. In a mixed equilibrium, the dispersion of beliefs is hump-shaped in l. Proposition 5 implies that, in a symmetric equilibrium, there is over-reaction at the individual level iff $l\alpha_1 < \frac{1}{2}$ and there will be under-reaction at consensus level if either the set of reasonable beliefs is small or if l is high. Moreover, both CG_i and CG_a are increasing in l. Proposition 6, implies that, in a mixed equilibrium, the dispersion of individual CG coefficients is hump-shaped in l.